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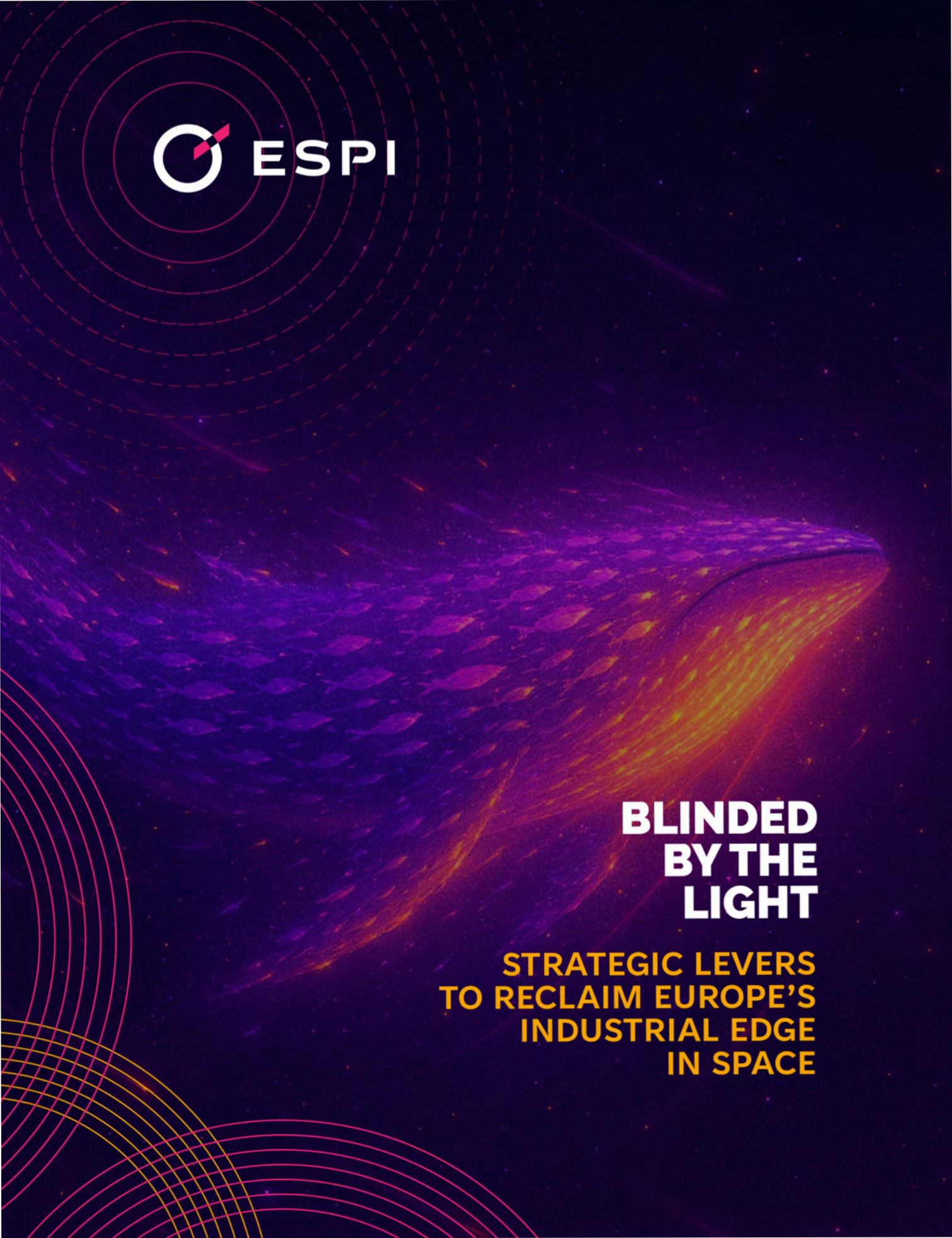
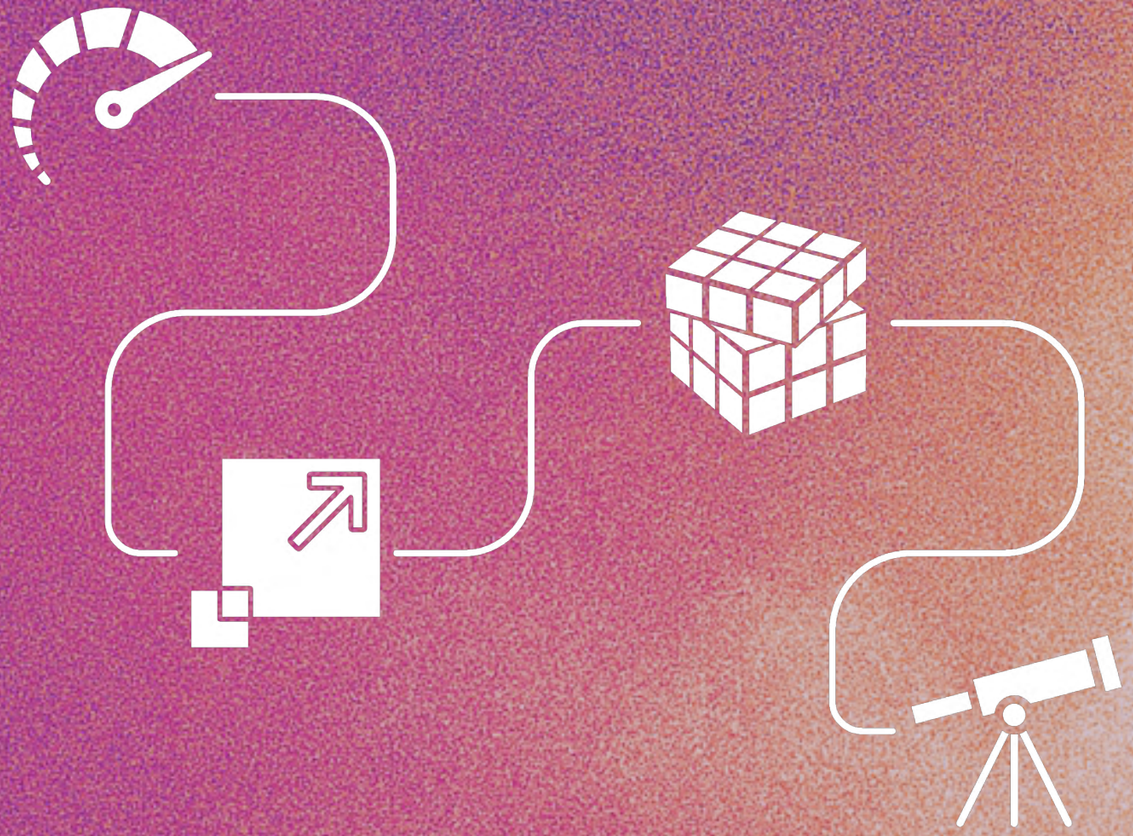


BLINDED BY THE LIGHT

STRATEGIC LEVERS TO RECLAIM EUROPE'S INDUSTRIAL EDGE IN SPACE



ABOUT

This report has been elaborated by ESPI as part of its mandate to provide independent analysis, proposals, and recommendations on strategic space issues to European decision makers and institutions. It expresses ESPI's **own views** stemming from its research and analysis and informed by two high-level workshops held in Luxembourg and Berlin in mid-2025, with **33 high-level participants** from industry, ESPI members, members of the ESPI Advisory Council and external experts, representing **the full value chain of the European space industry and with a wide European footprint** (see Page 64). Selected quotes from participants are featured throughout the document without attribution, and opinions expressed in this report do not represent individual views of any particular participant. Furthermore, the report considers related work of ESPI across its research agenda, such as its study on "The (Re)convergence of Europe's space and defence industries", "Space Venture 2024", the Brief on "IRIS[2] Growing Up: From Strategic Roots to Commercial Play", and "An Initial Perspective on a European Space Strategy 2040: From Vision Towards Implementation".

FOREWORD

Industrial capacity, financial markets, talent and skilled workforces provide the foundations for a **competitive European space ecosystem required to innovate** and to manufacture space infrastructures and to provide the services needed to fulfil the missions and purposes of space for Green and Sustainability, Security and Defence, and Exploration and Science. The digital economy, increasingly relying on space data and services, is becoming a critical component of national and European transformation. In this context, a growing European internal market, both institutional and commercial, is required to grow and maintain the European space industry and **to avoid overdependence** on foreign markets challenged by trade wars, the risks of takeover through foreign financing and acquisitions and the loss of Europe's talent through brain drain.

It is over a year ago now that the report on the future of European competitiveness, prepared by Mario Draghi to guide future EU strategy, recognised space as a key strategic sector that will define Europe's prosperity for decades to come.[1] Political debate at the highest level acknowledged the critical situation of Europe's space industry as a serious concern for the European economy. The **ESPI Advisory Council recommended addressing this issue**, underlining ESPI's added value as an independent Think-Tank acting also beyond national and individual industrial interests.

> *Competitiveness and sovereignty need each other, one cannot be sustainable without the other.*

This report has the objective to contribute to and widen the political debate on reclaiming Europe's industrial edge in space. It suggests looking at the **full value chain of space, from launchers and manufacturing to operations, data, and services,** to address all industrial stakeholders, from mature actors to new entrants, from start-ups to large industries. It puts **speed, agility and innovation at the core,** as most critical for success in a global space race. It acknowledges the ability to scale up as the crucial next criterion for success, with governmental and B2B markets as a catalyst. It is clear that beyond B2B and B2G, the ability to conquer consumer mass markets is central to survival as part of the digital economy. It emphasises the challenge of protecting Europe's knowledge assets and nurturing the next generation of talents.

The increasing role of security and defence in space is evident, attracting a lot of attention today, yet **long-term competitiveness will depend on scaling beyond governmental markets.** Emerging markets, such as direct-to-device connectivity, and simultaneous prospects of industry consolidation, potentially driven by actors outside the space sector, like mobile network operators, point in this direction. At the same time, growth might not be limited to emerging mega-constellations, a multitude of small- to mid-size constellations are also expected and already in development, each tailored to specific mission needs (e.g. automotive) and demanding more flexible and disruptive procurement and business models. **These procurement models go beyond the mere provision of hardware, addressing the full lifecycle of infrastructures and services and their evolution** over time as turnkey solutions.

Taken together, these trends underline that reclaiming industrial edge cannot come from simply scaling up the existing offer and processes. It requires the agility and high dynamics of innovation of the entire sector to conquer markets, governmental and B2B2C. **Competitiveness and sovereignty need each other, one cannot be sustainable without the other.**

Europe still has all it takes to reclaim its edge in space. This report urges the space community to elaborate an (industrial) vision, reclaiming space as an enabler of prosperity and peace, and aims to provide an impulse for more substantiated and structured debate towards such a vision, between institutional and industrial actors, between the space community and policy makers.

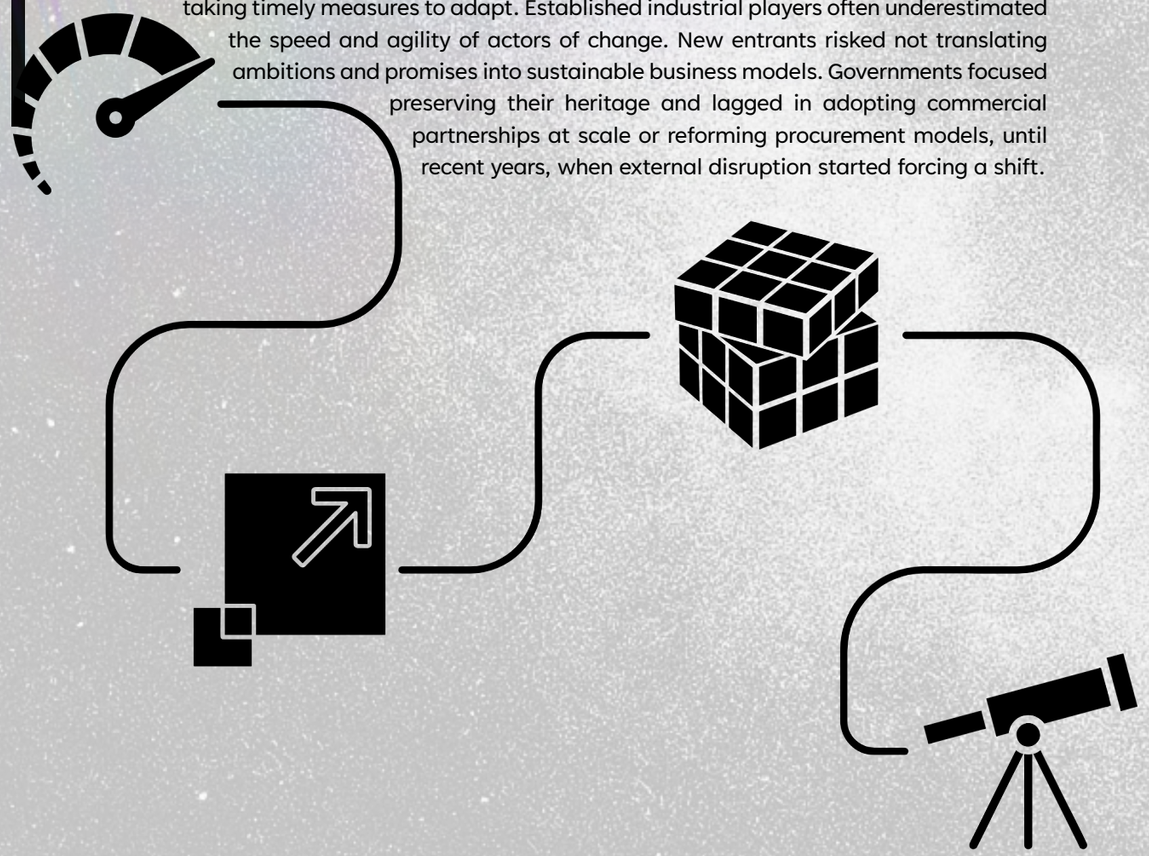
H. LUDWIG MOELLER
DIRECTOR, EUROPEAN SPACE POLICY INSTITUTE

[1] *European Space Policy Institute, ESPI Brief 69; The Draghi Report: It takes a village, September 2024*

HIGHLIGHTS

Europe in space is being blinded by the bright light of disruption of the global space ecosystem, brought by new industrial challengers like SpaceX, by rising space powers like China, and affecting even the most established institutional global actors like NASA and industries like Boeing.

For a long time, confidence in past success blinded Europe to the warning signs of global disruption and new technologies. **Niche competitors and emerging disruptors were often and for too long dismissed as irrelevant until they redefined markets.** The speed and intensity of disruption left leadership in government and industry without clear answers. A whole ecosystem of institutional actors and industry leaders, all used to decades of success in an environment dominated by government contracts, slow innovation cycles, and high-cost programmes, did not succeed in taking timely measures to adapt. Established industrial players often underestimated the speed and agility of actors of change. New entrants risked not translating ambitions and promises into sustainable business models. Governments focused preserving their heritage and lagged in adopting commercial partnerships at scale or reforming procurement models, until recent years, when external disruption started forcing a shift.



PUTTING UP A FIGHT

Europe's space Industry is losing market shares, struggling with profitability, and is challenged to keep pace with global innovation. Yet, unlike in many other DeepTech sectors, the balance of power is still ours to influence.

PROSPRITY & PEACE FOR FUTURE GENERATIONS

Only with a thriving and competitive space industry can Europe futureproof its defence, supercharge innovation, boost its champions in the digital economy and in globally competitive sectors such as automotive, agrifood, and pharma.

IT TAKES A VILLAGE

Space in Europe must be addressed through the prism of the full value chain, from the upstream manufacturing industry, satellite operators, to downstream services, and vertical integration into the wider economy. Mature players and new entrants. From Helsinki to Lisbon, from Warsaw to London.

BACK TO THE ROOTS

Europe must acknowledge that Competitiveness and Sovereignty need each other. Above all, sovereignty and increased defence expenditures should not be used as an excuse to justify non-competitiveness and governments fixing markets.

Read more on page 16

SPEED OF LIGHT

Speed and agility must be at the core of Europe's industrial revival. Without accelerating innovation pipelines, go-to-market strategies and turning them into a strategic advantage, our ambitions for autonomous security and reclaimed market shares are dead on arrival.

Read more on page 20

MODEL T

As space is embraced by digital hyperscalers and defence industries with sizable budgets and recurrent needs, the industry is challenged to move from artisanal processes to large scale industrialisation and serialisation. High risk, high reward.

INNOVATION IMPERATIVE

Skill is more than just the (valuable) engineering base. Skill means ensuring disruptive innovation is available to feed large-scale procurement, broadening the talent base to finance, business, policy, and PR, protecting strategic knowledge assets, and capitalising on top-tier global ranking in innovation.

Read more on page 40

FROM BIDDERS TO BUILDERS

While Europe has no shortage of declared ambition, its translation into capabilities continues to lag. A system-wide impact requires a shared vision that involves industry but also challenges it to think globally and disrupt.

Read more on page 50

BEYOND THE TEXTBOOK

The corporate world is unforgiving. Read ESPI's perspectives on major corporate developments in Europe and whether they are seen as moves protecting interests or spearheading innovation and global growth.

Read more on page 56

THE WINNING HAND

Europe cannot only rely on established processes or past success to secure its future in space. This report sets out actionable short- and mid-term recommendations across Speed, Scale, Skill, and Vision, to align ambition with execution and reclaim competitiveness and sovereignty.

Read more on page 60

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ESPI European Space Policy Institute

TAKING CONTROL OF THE LIGHT

Europe's Competitiveness and Innovation gaps are widening, and **space is no exception**. Once a global export leader across a spectrum of products and services, Europe is now facing a massive challenge in terms of receding market shares, low profitability and an ecosystem that is no longer able to keep pace with global disruptions and evolving needs.

But, unlike in many other DeepTech industries, the **balance of power is still ours to influence**, as long as Europe acts with bravery and intent. In doing so, it should not only rely on increased defence expenditures for space as a cushion offering short-term relief, but as a springboard for new innovative products, solutions, and services with an ambition to reclaim global markets.

> *In 5 years' time Europe's space industrial base will look very different.*

Space today is exposed to two much larger sectors: **defence and digital**. Ultimately, the European space industry can only thrive if it can enable both commercial and public benefits. It needs, in particular, to demonstrate its value in propelling the creation and growth of European hyperscalers in a global digital economy, and show governments how it can help ensure Europe's security and autonomy.

The war in Ukraine underscores how indispensable space-based capabilities are to modern warfare, where it's common practice for hundreds of space assets to be in sync, serving as a **force multiplier** in a single military operation.

The space sector spreads technological innovations throughout the economy. When paired with other sectors, such as automotive, transportation, agrifood, pharmaceuticals, and digital, **space supercharges innovation**, boosts the wider economy well beyond the space sector, and creates under-accounted value to society.[2]

> *Public procurement never asks* how European are you?

Sovereignty and competitiveness are not bought off the shelf, however. They can only be attained, sustained and expanded based on an innovative, dynamic, and market-driven **European industrial base** guided by long-term clarity, a strict customer benefit focus, and a vision on capability and infrastructure needs.

2 *European Space Policy Institute and Boston Consulting Group, More than a Space Programme: The Value of Space Exploration to Empower the Future of Europe, November 2023; National Institute of Standards and Technology, Economic Benefits of the Global Positioning System (GPS), June 2019.*

This risk is exacerbated by the EU-US Trade Deal of July 2025, under which Europe commits to purchasing €40 billion worth of US AI-enabled semiconductors, going even further, it submits to helping the US maintain its technological edge. If it is AI now, will it be space tomorrow?

The **European space industry is undergoing a profound transformation**, the most significant since its inception more than 50 years ago. This change highlights the vulnerabilities of its economic, industrial and political model, jeopardising the know-how, capabilities, technologies, and intellectual property developed over decades.

To build a future **free from crisis-driven reflexes**, Europe must build an industrial policy that spans the **full value chain** and reject short-term survival tactics. From upstream manufacturing to satellite operations to downstream services. It must leverage legacy players and new entrants. Build up procurement processes that simultaneously enable fast capability deployment, ensure a dynamic innovation pipeline, and, unlike today, provide incentives to improve performance and reduce cost once the contract is sealed.

Europe should also harness its geographic resilience, as industrial capacity is no longer concentrated. It stretches from Helsinki to Lisbon, Warsaw to London. This **diversity is a cornerstone of strength**, and more vital than ever in an era of intrusions into European airspace and continuous hybrid warfare.

Taking control of the light means recognising two dynamics. First, industry must develop ownership of challenges by shaping its own vision of innovative capabilities and services, including by taking more commercial risk. Second, public policy is challenged to create conditions to harness world-class know-how and turn it into a lasting strategic edge under European control, without enabling third-country subsidiaries to deepen the continent's dependencies.

The pace and degree of moving across the four strategic levers, **Speed, Scale, Skill,** and **Vision,** will determine whether Europe can claim its place as a global space power.

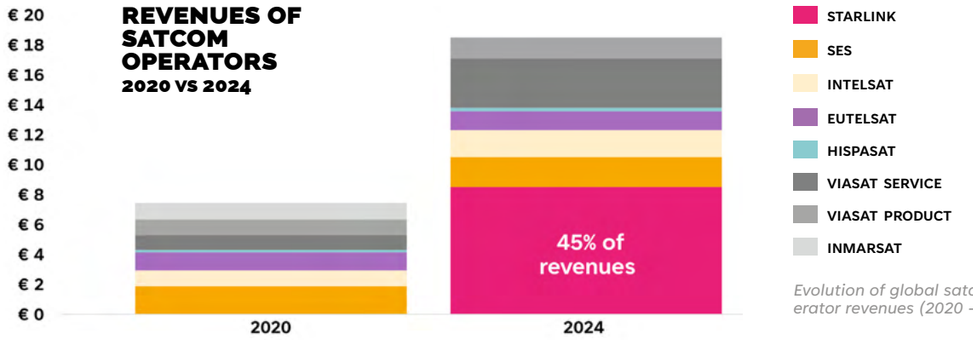
> *If there's no sovereignty, the market doesn't exist.*

> *If sovereignty is the only reason for space, our competitveness will suffer.*

RESET VS REPLAY:
PATHWAYS FOR EUROPE'S INDUSTRIAL ECOSYSTEM

For decades, Europe was considered a leading space power, demonstrating excellence and pioneering in both upstream and downstream segments. Despite relatively modest public budgets compared to other global players, Europe had built a competitive advantage by **focusing on innovation** and relying on a **unique framework for collaboration between institutional and commercial space** that made efficient use of public resources while leaning on commercial leadership.

This approach enabled Europe to benefit from discounted strategic autonomy in critical areas, as it gave rise to a highly competitive space industry that has long dominated global commercial space markets.



Evolution of global satcom operator revenues (2020 – 2025)[3]

Today however, the European space industry is undergoing a profound transformation driven by endogenous and exogenous forces.

The European industry, in particular established players, faces mounting pressure from shifting market trends in commercial markets, disruption by "Neoprimes" establishing themselves as new market leaders,

3 *Revenues of Intelsat-SES shown separately. Viasat revenue in 2024 reflects revenues post-Inmarsat merger.*

and geopolitical factors pushing governments worldwide to significantly increase their expenditures in order to consolidate their strategic leadership, thereby widening budgetary gaps even further.

Internally, the maturation and growth of the first European NewSpace wave has paved the way for the creation of new, and innovative actors. Still, the European space industry at large must **reinvent itself**, as it faces challenges in terms of profitability, bottlenecks to its supply chain, declining global market shares, a slow reaction by industry to under-accounted changes in user behaviour, moving from broadcast to broadband, and a slow reaction of public actors to the worsening security landscape.

In parallel, the Russian invasion of Ukraine and shifting U.S. foreign and trade policy under the Trump Administration are pushing Europe to (re)develop its defence industry, opening investment and market opportunities for the space sector..

However, while the projected increase in defence expenditures across Europe may ease the immediate pressure on executive leadership and provide a **temporary relief** for corporate margins, on the back of government deficits, it cannot alone carry the weight of long-term competitiveness of the space sector.

The moment should not be seen as a cushion, but as a catalyst to reimagine and transform Europe's space industry.

> *How did we go from being a continent that led in these domains to a moment where everyone doubts, and people ask whether decline has already set in?*
>
> *Sebastien Lecornu, 13 June 2025 (unofficial translation)*

> *You create disruption when you break the existing model*

RESET VS REPLAY:
PATHWAYS FOR EUROPE'S INDUSTRIAL ECOSYSTEM

COMFORT DULLS THE HUNT

European satellite operators in the 1980s and 1990s achieved growth and market dominance by successfully transforming the business model from specialised governmental and B2B connectivity to B2B2C for commercial markets. This was driven by significant innovation in SatCom Technology (satellites, antennas and modems), the cross-industry ecosystem collaboration, the agreement on common standards (e.g. Digital Video Broadcasting (DVB)) and the successful promotion of a Direct-to-Home (DTH) usage from GEO orbits. The resulting success and the move to privatise former intergovernmental (IGO) operators however had effects by the 2010s, ***shifting focus from transformation to consolidation, from innovation push to a focus on shareholder value****, harvesting mass market success to maximise short-term commercial profit. The long-term contract and revenue backlog (on average 10-15 years) created by the successful broadcast business attracted a different set of shareholders with low risk tolerance (e.g., pension funds) looking for long-term stable returns and favourable financial ratios.*

This shift happened to the detriment of disruptive innovation, no longer prioritised by the mature market actors. Risk-tolerant entrepreneurial new concepts and disruptive ventures or investments outside of the highly profitable core business often did not find approval at board level. "Times were good, and all should stay as is...".

By 2020, the disruption and innovation, which 20 years earlier provided the foundation for the commercial success of European operators, was introduced from outside Europe by new entrants with high-risk appetite and deep pockets. This was made possible, as ***the established European actors did not leverage their market position and did not respond to a slow but steady shift in user behaviour in the Internet age****. From Direct-to-Home (DTH) Broadcast to Direct-to-Consumer (DTC) Internet-type applications and with it a shift from GEO to LEO orbits. European operators still focused on risk and cost reduction, have only in recent years, forced by outside competition, started exploring ways to fully enter a new innovation cycle towards the 2030s.*

This new cycle now also includes new applications such as internet broadband, IoT, D2D, and integrated applications. Latest developments in Direct-to-Device (D2D) solutions, e.g., from Very Low Earth Orbit (VLEO), and the growing global demand for ubiquitous mobile connectivity represent a new level of push towards mass-market solutions, hyper-scaling and market dominance. This is not unlike the DTH offering, which stood at the origin of successful European operators decades ago. Speed and Spectrum *are two core factors of success. Without spectrum, there is no market access and without speed in launch and commercialisation, there is no viable business model. In GEO-times, operators had the luxury to be break-even in 3 to 5 years and could then "home-in" another 10+ years of attractive returns. In the new LEO-world, while launch costs have come down, the number of satellites and related CAPEX for a globally competitive fleet has grown, innovation cycles and lifetimes are shorter, and returns need to flow in faster for the fleet upgrades to be viable.*

In this context, the role of the IRIS² programme is critical, beyond providing governmental solutions, to also act as a catalyst and funding backbone for the development of innovative commercial solutions, including B2C applications (such as D2D and Direct-to-Vehicle) to enable the economies of scale required to remain competitive on a global market.

The solutions and service developed within the IRIS² framework need to become not only a success within Europe – like DTH in the past – but require a global ambition, enabling European space Industry to grow markets and customers worldwide.

A SPACE POWER OF THE PAST - A SPACE POWER OF THE FUTURE?

This brings Europe's space industry to an inflection point, driven by mounting challenges, but also by new opportunities, with the defence and digital sectors showing a new level of interest in space, providing a unique window of opportunity to the full space industrial complex to innovate and scale.

As space returns to the forefront of global geopolitical and geoeconomic ambitions, massive investments in the United States and China are already enabling their space industries to benefit from large-scale programmes, based on which they can expand commercially, undermining the market positions of the European industry.

For now, Europe's response is slow, stepwise, and siloed; treating the launcher crisis, the economic viability of its manufacturing base, and the need for unrestricted access to secure satcom one by one, but not as part of an **overall policy and industrial strategy**.

WHEN EUROPE MAKES A BOLD MOVE

"U.S. dominance of free world space transportation faces strong competition. [...] The Europeans and the Japanese have targeted specific space technologies for development, and they will soon be providing stiff competition for services heretofore offered only by the United States."

U.S. Office of technology Assessment (1982)

Europe's **past discount on strategic autonomy now comes at a cost**. At a time of geopolitical shifts and trade wars, Europe's underinvestment in security, defence, and resilience now represents a major handicap, as global public investment in space is overtaking civilian space budgets; Europe's gap in strategic capabilities is widening fast.

Compounding this handicap is Europe's fixation on normative behaviour, rule-setting, and governance, without building the capabilities that give those efforts force and legitimacy.

Europe is faced with a perfect storm. This is the moment to prove its strengths, re-focus its vision, and act.

> *We should, once and for all, become completely independent from the United States with no turning back.*

The continent has all it takes to reposition itself as a **force of innovation**, develop best-in-class sovereign capabilities, as proven with Galileo, and nurture a thriving commercial base fit to win global markets in a digital economy.

Even today, Europe remains the **supplier of choice for high-end space systems** such as state-of-the-art Earth observation systems, pressurised space station modules, ground station services, as well as high-performance enabling components such as laser communication terminals, dispenser systems, or electric propulsion.

A SPACE POWER OF THE PAST - A SPACE POWER OF THE FUTURE?

These examples, alongside many others, prove that Europe's space sector maintains world-class technical excellence, and contains seeds of entrepreneurial spirit throughout the value chain. Paired with a strong, publicly backed innovation ecosystem, addressing key future technologies as for example laid out in the ESA Technology 2040 Vision, Europe can reposition itself as a pioneer in new technologies and derived applications and services.

This is further reinforced by a host of **existing strengths** that hold the potential to propel Europe towards the status of a full-fledged space power.

EUROPE'S TOP-TIER PERFORMERS IN THE GLOBAL INNOVATION INDEX
(SOURCE: WIPO, 2025)

1. Switzerland	6. United Kingdom
2. Sweden	7. Finland
3. United States	8. Netherlands
4. Republic of Korea	9. Denmark
5. Singapore	10. China

TO BE A SPACE POWER:
EUROPE'S EXISTING SOURCES OF STRENGTH

An **innovative ecosystem of new companies** taking early leadership in their market segments emerged – some now scaling towards midcap status

Elevation of space on the European political agendas, affirming the sector's essential role in Europe's economic growth and defence

A **rich industrial heritage** and space systems for space science and services in key applications (climate monitoring, PNT)

Increased and continued interest of **private investment** showing belief in **long-term viability** of the European space sector over the past decade

Recognition as a reliable partner in **international collaboration** opening up potential for increased diplomatic gains, exports, and global market penetration

A **ready-to-use procurement toolbox** fit to drive commercial developments (e.g. ARTES)[4] and new models in the process of implementation to maximise the efficiency and innovation (e.g. ELC, CCM, CCTI)[5]

4 *ESA's programme of Advanced Research in Telecommunications Systems*
5 *ELC – European Launcher Challenge; CCM – Copernicus Contributing Missions; CCTI – Commercial Cargo Transportation Initiative*

WHERE WE STAND: GEO satellite market decline, once the cornerstone of profitability for Europe's manufacturers

EFFECTS: Challenges to the profitability of manufacturers, falling market shares of European operators, industry layoffs

WHERE WE STAND: Lack of readily available European launch vehicles in line with evolving market needs

EFFECTS: Lack of launch capacity to support announced European ambitions at least for the next 5 years

WHERE WE STAND: Dependency on foreign commercial companies for high revisit, highly reactive, EO data sets

EFFECTS: Risks to be barred from national security intelligence based on third-country political decisions

WHERE WE STAND: Primacy of U.S. digital giants for ground segment infrastructure as the sector moves to GsaaS using cloud environments

EFFECTS: Even when the space infrastructure is made in Europe the brain power of European satellites is subject to foreign control

WHERE WE STAND: Limited (national) space-based missile tracking capabilities and scarce global coverage for incoming threats

EFFECTS: Europe-wide vulnerability to incoming objects, as shown in the uncontrolled Falcon 9 debris reentry over Poland

WHERE WE STAND: An underscaled space surveillance system falling short of offering global, persistent and high-fidelity domain awareness

EFFECTS: Compromised Europe's orbital threat assessment and limits its ability to shape global norms on space safety and sustainability

SOVEREIGNTY x COMPETITIVENESS: SOLVING THE EQUATION FOR A STRATEGIC ADVANTAGE

While institutional markets remain by far the largest source of revenue for the space industry, **industrial competitiveness is the only key to sustainable and long-lasting sovereignty**. The pace of innovation in competitive environments outpaces that of traditional institutional procurement programmes with rigidly defined specifications and limited incentive to improve performance and cost once the contract is sealed.

> *Without the innovative power, research, development, and production capacity of industry, we will achieve nothing in space [...] Let me be very clear: we in politics want you [industry] to succeed. Because your success means security for our country.*
>
> *Boris Pistorius, 25 September, 2025 (unofficial translation)*

Sovereignty itself is also critical to enabling further industry growth and expansion. Public ambitions, backed by clear needs and commitments, including anchor customer schemes, form the backbone on which companies can build to scale up and commercially expand.

In Europe, sovereignty and competitiveness are too often **wrongly presented as separate pursuits**, for example, questioning *"what is this launcher for? Is it to make life easy for commercial satellite operators or is it to assure European launcher autonomy"* in the development of Ariane 6.[6] It is essential that public programmes not only ensure autonomy, but simultaneously create opportunities for competition, ensure innovation and keep Europe's technological edge in an increasingly competitive global sector.

When discussing future pathways for Europe's space sector, many **call for a dedicated industrial policy for space**. A tailored approach is needed as space is a unique sector at the intersection of public purpose and private gain. Moreover, space services are increasingly integrated into broader domains such as energy, transport, digital, agriculture, next to defence and climate, no different to AI and energy storage technologies. Their trajectory will increasingly define the competitiveness and strategic posture of Europe as a whole.

> *Europe tries to create champions through budget allocation and not competition.*

A MARRIAGE OF COMMERCIAL AMBITION AND AFFIRMED SOVEREIGNTY

As early as 1960, European leaders, recognised the need for European industry to "participate in the commercial developments resulting from space technology" (CoE, Recommendation 251). In 1961, Lord Thorneycroft highlighted that the creation of ELDO was crucial to "avoid a situation in which Europe is right out of the launching of any satellite in the future," describing it as "an opportunity, which otherwise would have been denied to us, of taking some part in the commercial exploitation of space".

The well-known Symphonie case in the 1970s illustrated this in practice. First initiated through a political commitment between President de Gaulle and Chancellor Adenauer, the Symphonie satellites were only allowed to launch aboard U.S. rockets on the condition that Europe refrains from commercially exploiting them. This galvanised Europe's efforts to develop its own launcher, asserting its sovereignty in space access and for decades succeeding on commercial launch and satellite broadcast markets globally.

Today, Europe may be living its 2nd Symphonie moment.

For such an industrial policy to be effective, it must **address the full value chain**, from upstream manufacturing to downstream applications, including the end-user segment. This framing, now echoed in the European Commission's Vision for the European Space Economy, is essential not only to preserve industrial capabilities but also to expand demand, stimulate innovation, and ensure long-term strategic relevance.

> *Sovereignty should not be used as an excuse to justify non-competitiveness*

When charting a path forward, Europe must reflect on the foundations of its past success. **The origins of Europe's space programmes**, and the positive trajectory following the Symphonie case, offer important lessons, particularly the lucid recognition that autonomy and commercial opportunities reinforce one another and are key to an innovative and competitive space industry. Reclaiming this relationship will be key.

IRIS²: ENSURING AUTONOMOUS SECURE CONNECTIVITY AND ENABLING A COMMERCIAL EXPANSION

The success of IRIS² will depend on a firm commitment, by both institutional and industry actors, to its dual objective: providing both sovereign secure satellite communications and fostering commercial markets.

With the ambition not to replicate a purely publicly funded programme IRIS² should power both government connectivity, while enabling commercial broadband access globally. This broad objective underscores the programme's role as both the continent's institutional baseline infrastructure, as well as a catalyst to develop a more competitive, and innovative European SatCom ecosystem, including promising D2D and IoT markets.

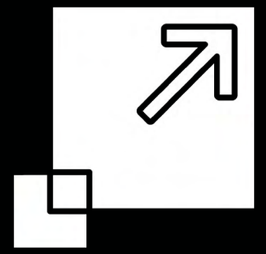
IRIS² should not be limited to the procurement of a strategic public asset, but also regarded as an opportunity for Europe's industry to develop disruptive commercial services. Only by achieving both can it realise its transformative potential.

6 *SpaceNews, DLR's Woerner Remains Unconvinced Just-unveiled Ariane 6 Design Is Right Way To Go, July 2013*

SPEED

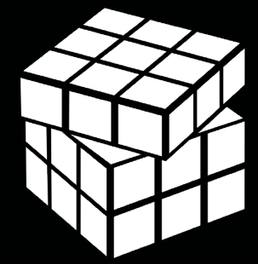
The agility and responsiveness of industrial action, programmes and policy frameworks, institutional support systems, and supply chains in adapting to market changes, technological advances, and competitive pressures.

SCALE



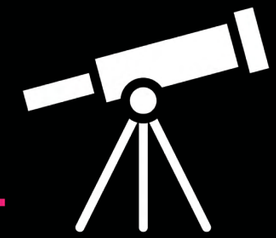
The conditions and ability to expand production, develop economies of scale, and extend market reach effectively across existing and into new market segments.

SKILL



The development and enhancement of specialised knowledge in industry and government, end-user competencies, and control over critical resources critical to driving innovation, industrial upgrading, and productivity growth.

VISION



The strategic clarity, long-term corporate & political direction, and forward-looking leadership necessary for guiding industrial transformation, ensure autonomy, and sustainable growth.

TO BOOST SOVEREIGNTY, RECLAIM MARKETS AND POWER EUROPE'S RENEWAL.

BLINDED BY THE LIGHT

STRATEGIC LEVERS TO RECLAIM EUROPE'S INDUSTRIAL EDGE IN SPACE

On 14 February 1876, both Alexander G. Bell and Elisha Gray submitted patent applications for the telephone on the same day. Bell's application was filed first, just hours before Gray's.

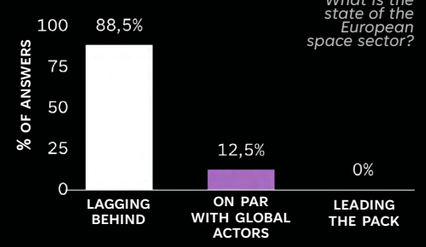
Space, like any other DeepTech industry, depends on bringing innovation to market quickly. Staying innovative means staying ahead of competition and ultimately becomes a question of corporate survival. **Now, as new market segments open and existing segments get disrupted, speed matters more than ever.**

The growing capital intensity of space business, combined with shortened innovation cycles, makes the stakes very high for any company entering this market. The result is a high-risk, high-reward game in which companies and investors are increasingly inclined to adopt a winner-takes-all strategy. In essence, **the window to lead is narrow.**

ASSESSMENT OF THE EUROPEAN SECTOR IN THE GLOBAL CONTEXT

SOURCE: ESPI WORKSHOP

Question posed: What is the state of the European space sector?



SLOW R&D TIMELINES FAIL TO PRODUCE REAL INNOVATION

We are at least 50% slower in public R&D and co-funded schemes compared to self-financing R&D.

You stay away from public R&D when you want a competitive edge.

When development cycles drag, chances to innovate and disrupt reduce. Today's public (co)funding programmes are designed to accelerate breakthroughs but often slow them down instead. While generally beneficial, firms tend to look for capital elsewhere when true market gains are on the line.

Public schemes should fast track innovation, guiding promising technologies from concept to market-ready products in a matter of months. A big share of Europe's R&D landscape, epitomised by Horizon Europe, is focused on delivering innovation through grants, often translating into bluesky workshops or requirement-laden programmes that do not deliver products to market. The European space sector needs a **full stack process** that fast tracks research into technology (grants), and technology into a product (procurement).

Public schemes should fast track innovation, guiding promising technologies from concept to market-ready products in a matter of months.

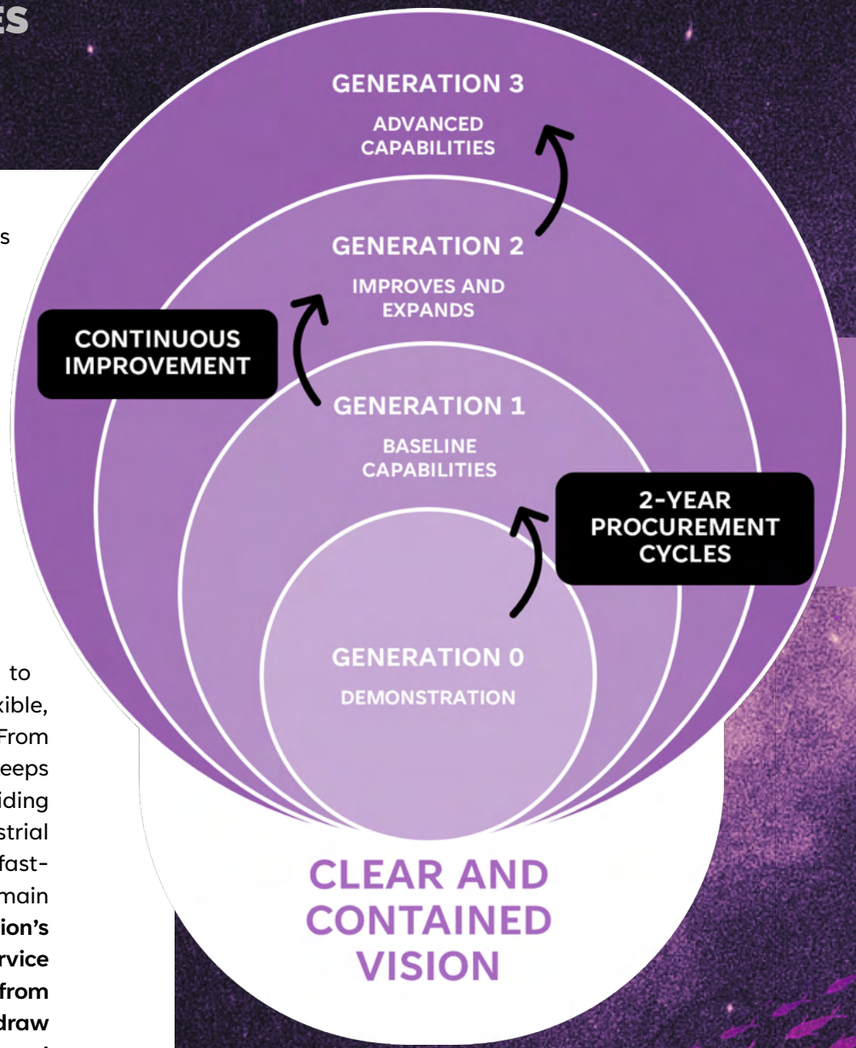
The US Space Development Agency (SDA) offers a compelling example of a paradigm shift in innovation processes, driven by the ambition to overturn the traditional sluggishness of public procurement in US defence programmes. This philosophy is based on the principle of rapidly developing cutting-edge capabilities to maintain a technological advantage. Its flagship initiative, the Proliferated Warfighter Space Architecture (PWSA), is based on a multi-layer space capability deployment plan that relies on agile and accelerated development cycles. The constellation is being built with a sharp and contained focus: to deliver beyond-line-of-sight targeting from space and detect and track hypersonic and advanced missile threats. Therefore, all the **innovation support by the SDA is driven and contained by these strategic objectives** in order to not overextend nor overlap the resources with other agencies.

The PWSA employs a spiral development model, where the constellation is deployed every two years through tranches to continuously introduce new technology and capabilities.

SLOW R&D TIMELINES FAIL TO PRODUCE REAL INNOVATION

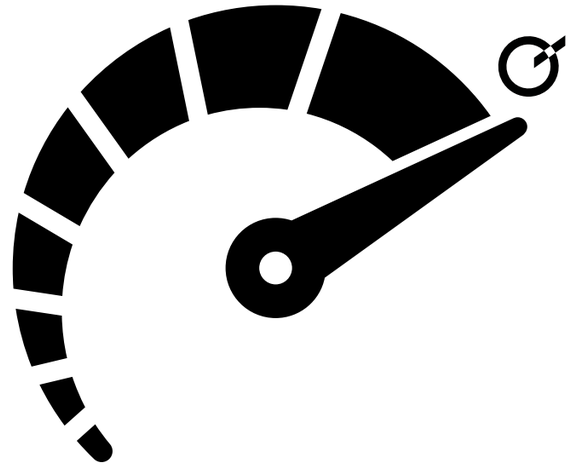
Importantly, the programme is not tied to a single company but spread throughout the industry, allowing intra-competition to deliver the best technology. Moreover, the programme has a specific on-ramp for emerging companies with low Technology Readiness Level (TRL) solutions called Hybrid Acquisition. This allows the SDA not to be vendor-locked to a small group of higher performing companies.

These characteristics are essential to make the programme lean and flexible, keeping the innovation edge. From the government's perspective, it keeps its capabilities up to date, providing enduring sovereignty. From the industrial perspective, it provides a pipeline for fast-tracking innovation, helping it to remain competitive. **The European Commission's Earth Observation Government Service and ESA's European Resilience from Space architectures should draw inspiration from SDA experiments and lessons learned in procuring innovation, and seize a unique opportunity for Europe to start establishing similar models of agile procurement.**



Spiral development used in the PWSA programme

> *The danger lies in treating new entrants solely as subcontractors, rather than reserving a share of prime contracts for new players in programmes like Copernicus, Galileo, or IRIS²*



NO LONGER A FIXATION: SPEED AS A MISSION-CRITICAL CONDITION

With the increasing demand for strategic space capabilities, by both military and civil security actors, speed is not only theoretical but a question of strategic survival. This need pushes the industry to offer almost immediate deployment and service delivery, as the profile of missions and needs of security actors are evolving towards rapidly deployable and reactive systems whereby **waiting to deploy a new asset for years is no longer an option**.

> *Speed requires the ability to take risk, and the European procurement model prevents any risk-taking.*

In modern warfare, no military operation can succeed without support from space systems. The European space sector faces a unique opportunity; design ambitious R&D programmes relying on clear procurement pathways turning innovation into operational assets that can shift the balance in a conflict.

NO LONGER A FIXATION: SPEED AS A MISSION-CRITICAL CONDITION

WHEN SPEED PAYS OFF

*On 23 June 2025 **ICEYE** launched 6 satellites aboard a Falcon 9 rideshare mission. The launch included ICEYE's first (out of four) 25cm resolution SAR satellites for the **Royal Netherlands Air Force** (RNLAF). The satellite was launched within **four months** of signing the agreement. Days after the launch, ICEYE announced a new Letter of Intent with the Finnish Ministry of Defence that enables the Finnish Defence Forces to acquire ICEYE SAR satellites.*

> *The satellite was launched within four months of signing the agreement.*

*Speed also pays off in civil programmes. **OHB Sweden** launched the Arctic Weather Satellite (AWS) in August 2024, three years after receiving a €32.5 million ESA contract under firm conditions: fixed cost, rapid delivery, and performance-based evaluation. After on-schedule delivery and almost a full year of performance completed, the EUMETSAT council has approved a **follow-on constellation**: EPS-Sterna in July 2025, comprising up to 20 satellites with a full contract value of approx. €850 million for the full system between 2025 & 2042.*

Further emphasising the importance of speed, a growing array of civil security crises and hybrid threats, from natural disasters to cyberattacks on critical infrastructure, is multiplying the demand for responsive space-based services, notably for intelligence and connectivity. This reality reinforces the need to develop an industrial model that can **deliver at speed** and satisfy the increasingly complex and multifaceted needs of actors with boots on the grounds as their operations unfold.

> *Industry cannot wait for institutional certainty before beginning development*

However, to satisfy these needs on time, and develop a strategic edge, industry cannot simply wait for institutional certainty. It must proactively scout technology, anticipate market trends, and take risks. If industry only begins development once public procurement systems are ready, the competitive advantage will be lost and the reliance on foreign suppliers inevitable, further weakening Europe's strategic capabilities.

ANTICIPATING DEMAND: GETTING EUROPEAN AUTONOMY AND COMPETITIVENESS OFF THE GROUND

Europe's only heavy lift launch vehicle, Ariane 6, has a backlog of over 30 launches, with strong demand for launch services observed globally. The 30 launches however do not include Europe's approved and proposed autonomy-driven systems (e.g. IRIS², ERS, and LEO-PNT), national assets (e.g. PAZ-2, SatComBw 3, Poland's VHR Satellites), nor the commercial expansion for projects like OneWeb Gen 2.

The challenged availability of launch vehicles is a result of constraints to ramp up production beyond the 10-per-year cadence currently foreseen for Ariane 6. As an illustration of current limits: the single ship, Canopée, transporting Ariane 6 stages between Europe and French Guiana is able to make the one-way transatlantic crossing in 10-12 days, capping its potential to about a dozen return trips per year, considering the need for downtime, maintenance and loading.

Should demand dynamics continue, while the cadence of flight does not increase by 2030, this might lead to three non-exclusive scenarios, each potentially detrimental to Europe's autonomy and competitiveness:

1. *A reorientation of European payloads to foreign launch service providers, exposing system vulnerabilities and supporting non-European market growth;*
2. *Enforced priority for institutional payloads, to the detriment of commercial customers, as once foreseen in the 1980 Declaration Relating to the Ariane Launcher Production Phase;*
3. *A delay or downsizing of key European space programmes due to constrained launch availability.*

This shows the importance of thinking of Europe's space ambition, and the speed of its attainment across the full value chain, not segmented across market verticals and single programmes.

BLINDED BY THE LIGHT
Strategic Levers to Reclaim
Europe's Industrial Edge in Space

INDUSTRIALISATION TO IMPROVE THE TIME TO DELIVERY AND AVAILABILITY OF SUPPLY

In a fast-paced market, the need for fast delivery and continuous availability of products and services is crucial, from satellites and launch slots, to user terminals and data. This requires the space industry to rethink its production rates and consider stockpiling in order to satisfy emerging and often time-restricted needs. Compared to a unit-by-unit approach, industrialisation and serialisation of the manufacturing process brought about by constellation architectures bring down the per-unit cost, under the condition of sufficient and sustained demand. This means significant budget announcements, such as Germany's €35 billion investment into space defence, need to be translated into capability plans and needs soon after announcement.

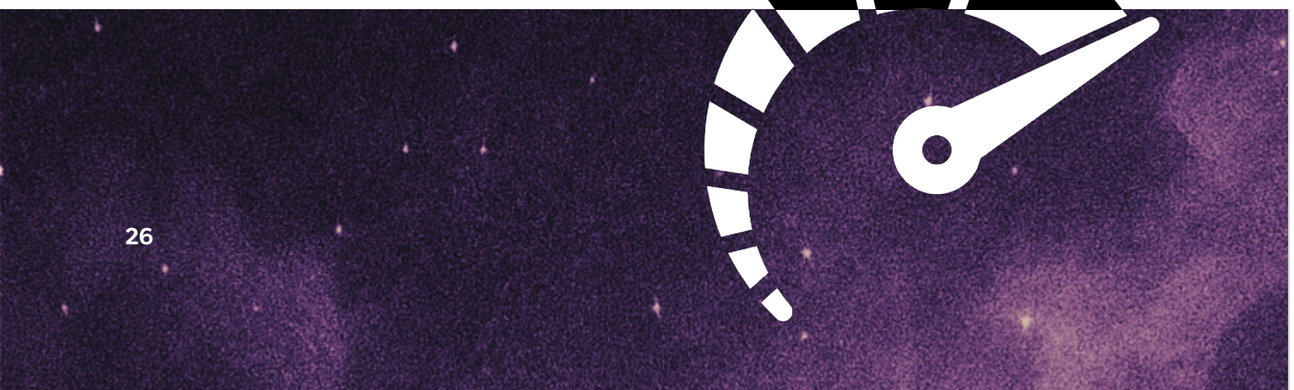
> *Clearly, today we see that the market, especially the institutional market, is growing faster than the supply.*
>
> *David Cavaiollès, March 2025*

SCALING SATELLITES MANUFACTURING: MAPPING OF CURRENT AND PLANNED EUROPEAN SATELITES

Several facilities have been built or have started construction in Europe over the past 10 years. All of them include elements of serialisation enabling production at scale of satellites in the 50-500kg range. Two examples are the Thales Alenia Space Smart Factory in Rome, completed during the Autumn of 2025, incorporating visualisation and simulation technologies, connected machinery, and process automation, and Aerospacelab's new 17,500 m^2 Charleroi facility, which aims to deliver 500 satellites yearly.

While European manufacturing capabilities remain far from the US or China's in aggregate, industrial actors seem determined to close that gap.

(see map figure on p.29)



ESTABLISHED PRODUCT VS. FRONTIER TECHNOLOGY

SODERN'S experience with its Auriga star tracker shows how a pre-built inventory, a lean process, and a one-size-fits-all approach enables successful serialisation of mastered technologies. Originally developed to meet a 1,800-unit order from OneWeb in 2016, Sodern delivered 1,350 units by 2021 after the constellation was downsized. Rather than scale back, the company kept its production line active and focused on the projected demand, offering its Auriga tracker based on the OneWeb design. By 2024, more than 1,250 satellites had flown with Auriga trackers. As of 2024, Sodern now delivers around 400 units per year with lead times of less than one month.

MYNARIC'S story highlights the structural challenges of transitioning from technology development to serial production. After early technical capability and major contracts, including multiple awards under the U.S. SDA Tranche 1 programmes, the company aimed to scale output from 11 terminals in 2020, and 50 in 2021, to a projected 2,000 units per year. However, the industrial ramp of its CONDOR Mk3 terminal was hindered by challenges in production yields and supply chain reliability. These issues resulted in shipment delays, growing backlogs (from 256 units in 2022 to 794 in 2023), and ultimately a financial restructuring in 2024, leading to an announced acquisition and integration by Rocket Lab.

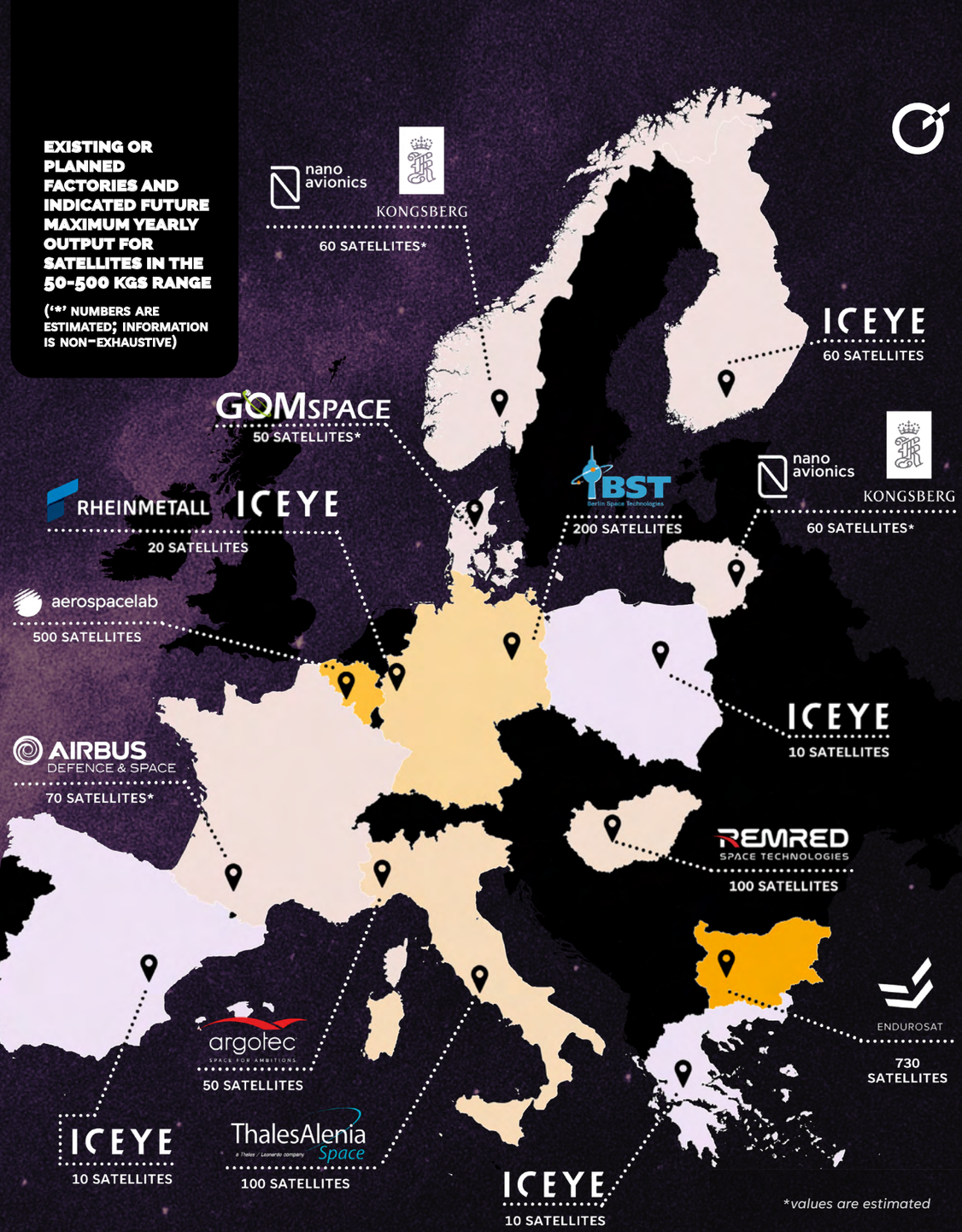
The **need for industrialisation** has been recognised by industry executives and shareholders alike. Space companies are addressing this by importing processes and staff from other industries and by investing in new manufacturing facilities designed specifically to scale production of small and medium-class satellites and components. Success in these efforts ultimately depends not only on strong market demand, but also on stable, resilient supply chains capable of delivering on-demand and withstanding supply shocks.

INDUSTRIALISATION TO IMPROVE THE TIME TO DELIVERY AND AVAILABILITY OF SUPPLY

The industrialisation of space products, at systems and subsystems levels, is however no easy feat even when demand and infrastructure are in place. Regardless of product technical maturity or corporate heritage, industrialising from dozens of units per year to serial production entails significant business risks, sustained investments and transformation of internal operational processes. This challenge becomes even more acute when dealing with cutting-edge technologies, i.e that push the boundaries and lack established industrial pathways.

> *When the design is physically located close to manufacturing, the design can be improved to help the manufacturing produce at speed and scale. This is one of SpaceX and Tesla's ingredients for success.*

As competition continues to intensify across the space value chain, and as industry players consider deeper vertical and horizontal integration, not every company will see returns on their investment. As in any commercial market, some will need to pivot, tie-up, or face bankruptcy. There is however a lot to win in an environment where responsiveness is increasingly valued, as winners might ultimately not be determined by marginal gains in performance, but by the **ability to deliver their qualified products and services on time and at the required frequency.**



SLUGGISH CAPITAL ROLLOUT PUTS OUR INDUSTRY IN AN UPHILL FIGHT

In Europe, access to finance, both in the form of grants and investment, is generally slower than in the United States across the DeepTech landscape. Sluggish deployment of capital and lack of fast-track processes for institutional procurement undermine the speed at which companies can invest in industrialising or commercialising new offerings. As a result, **European companies are at risk of not having the financial resources to iterate as quickly as their competitors** to create technological edge, resulting in difficult-to-overcome technology and market-share moats.

SPEED KPIs IN MAJOR GRANT SCHEMES IN EUROPE
AND ITS PERFORMANCE SINCE 2022 ROUNDED BY THE UNIT

SPEED KPIs	EUROPE			UNITED STATES	
	EIC PATHFINDER	EIC TRANSITION	EIC ACCELERATOR (GRANTS)	SBIR PHASE I	SBIR PHASE II
TRL	1-4	4-7	6-9	1-3	4-6
TIME TO INFORM (KPI)	5	4	4	3	3
TIME TO INFORM (ACTUAL)	4	5	4	3	3
TIME TO GRANT (KPI)	8	6	8 (6 SINCE 2024)	6	6
TIME TO GRANT (ACTUAL)	6	9	9	5	6

ALL KPIs MEASURED IN MONTHS • TIME TO GRANT INCLUDES TIME TO INFORM

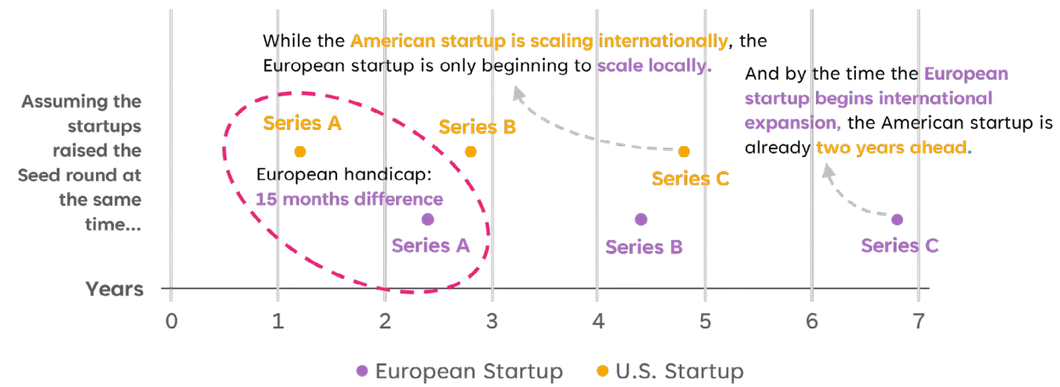
Comparing two major grant instruments in Europe and the United States, the European Innovation Council and the Small Business Innovation Research programme (SBIR), highlights the issue. Both use two key performance indicators related to speed: (i) time to inform, that is the period between the closure of a call for proposals and the date when applicants are told the results, and (ii) time to grant, meaning the period between the closure of the call and the signing of the grant agreement.

The first difference is that **EIC targets are less ambitious. As a baseline**, U.S. institutions aim to inform companies sooner, which can be decisive for whether to pursue a product or service or pivot. When it comes to the time it takes to sign a grant agreement, SBIR again sets a more ambitious and consistent target. In 2024, the EIC Accelerator shortened its target from 8 to 6 months, but the EIC Pathfinder still retains the 8-month objective. The second difference is that **performance in the European programme has not met the defined KPIs:** SBIR has generally met its targets, while within the EIC only the Pathfinder has consistently done so, and even then, against less ambitious timelines.

This is only a snapshot of the grant ecosystem in Europe, where in many programmes it is not clear if similar "speed metrics" are even defined. With the EIC being one of the major European financing mechanisms, it is indicative of a broader problem.

> *Institutional players should be buying & contracting innovation, not investing or subsidising it.*

Another key difference between the EIC and SBIR programmes is that only the U.S. scheme has the explicit goal of building an innovation pipeline for the U.S. government agencies, particularly the Department of Defence (DoD), with a clear product in mind. **This clear path to commercialisation benefits both sides.** For the government, it drives private R&D while giving the DoD the option to quickly procure new technologies. For companies, it offers more than a grant, potentially **providing fast-track access to an institutional customer and revenue stream that can unlock further and faster private investment.** SBIR Phase III illustrates this, as it is **directly funded through procurement contracts** rather than the programme itself. European grants, often define more abstract objectives (e.g. defining the architecture of a joint system), leading to a higher risk of creating a path to nowhere.



The sluggishness of capital, partly also driven by the above dynamics, extends to (private) investment. **European space companies are much slower raising capital than their U.S. peers.** This is particularly acute in early stages, where European investors are more present, startups take more than twice as long to progress from Seed to Series A.[7]

The lag compounds at later stages. By the time a European company reaches Series B, it would need to raise its next round in just six months to catch up, while an American competitor has had almost two years to focus on building production capacity, fulfilling mostly domestic or pilot contracts, and lining up first international customers rather than pursuing full-scale foreign market penetration.

7 European Space Policy Institute, Space Venture 2024: Global Investment Dynamics, June 2025

BLINDED BY THE LIGHT

STRATEGIC LEVERS TO RECLAIM EUROPE'S INDUSTRIAL EDGE IN SPACE

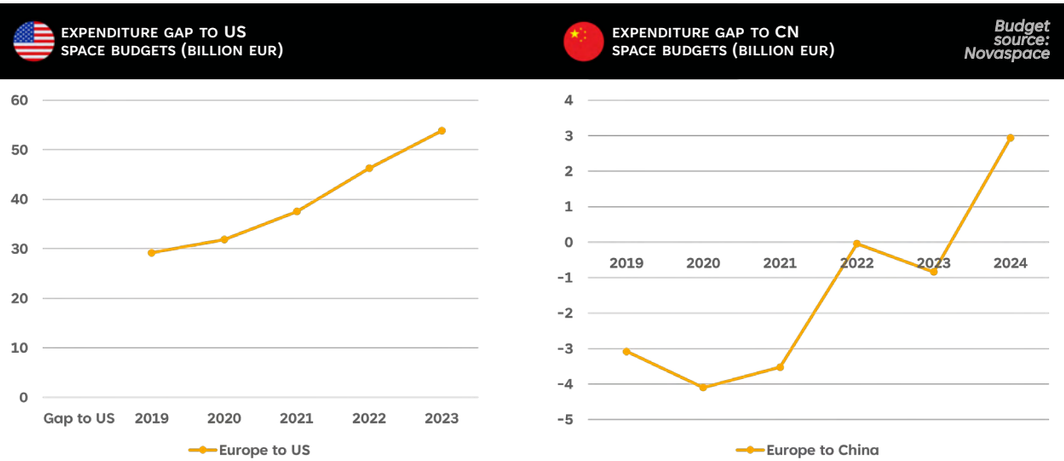
SCALE

> *Public investment in space R&D in the EU does not meet the required level of ambition.*
>
> *Mario Draghi, The future of Europe's Competitiveness (2024)*

Scale is not just a question of size and growth; it enables systemic change and global leadership. The European space sector struggles to achieve significant scale, hampered by a fragmented domestic market, artificially low profitability, risk aversion and a recurring lack of ambition. As space becomes an integral part of the digital economy and defence readiness, Europe must overcome its structural barriers by stimulating both public and private investment, reforming access to public markets, and ultimately fostering an industrial ecosystem where competition drives future innovations.

BILLION EURO GAPS RESULT IN TRILLION EURO LOSSES OVER A DECADE

The gap between public spending on space in Europe and that of the world's leading space powers is widening at an alarming rate. The cumulative gap between European and the U.S. space spending is worsening annually at a rate of 15%, **doubling since 2019 from EUR 29 billion to EUR 58 billion**. If the funding gap with the U.S. continues to grow at the current pace, **Europe could lose the equivalent of over USD 780 billion by 2034, a colossal loss in a sector still largely driven by public infrastructure investment.** China surpassed Europe's space expenditures in 2024 by EUR 3 billion, further challenging European leadership and competitiveness.



The disparity between European and U.S. public spending in space cannot be justified by a structural gap in overall public R&D investment. **European public spending on R&D is roughly comparable to that of the U.S.; however, when it comes to space, the U.S. outspends Europe by a factor of five.** It is therefore mainly a question of ambitions and priorities. The situation becomes even more problematic in the military space domain as the gap with U.S. expenditures jumps to 15x. Only 15% of Europe's public space budgets are earmarked for military space activities, while the global average is about 50% and further increasing.

Without co-investment and procurement support of adequate scale, European space companies struggle to reach growth stages, leading them to seek funding overseas to compensate for low budgets, relocating to larger markets, and ultimately undermining autonomy in

> *The Golden Dome decided by Trump will bring a new source of financing that dwarfs anything in the commercial upstream sector.*

BILLION EURO GAPS RESULT IN TRILLION EURO LOSSES OVER A DECADE

strategic capabilities. Europe could draw inspiration from large-scale programmes implemented in the U.S. to develop strategic capabilities and serve as an anchor client for space companies. A prime example is the recent CHIPS and Science Act, through which the US government is investing over EUR46 billion, efficiently relocating semiconductor supply chains back to the U.S.

> *Our challenge is how we can scale without the same public budget. If we double, triple the budget the difference is still huge.*

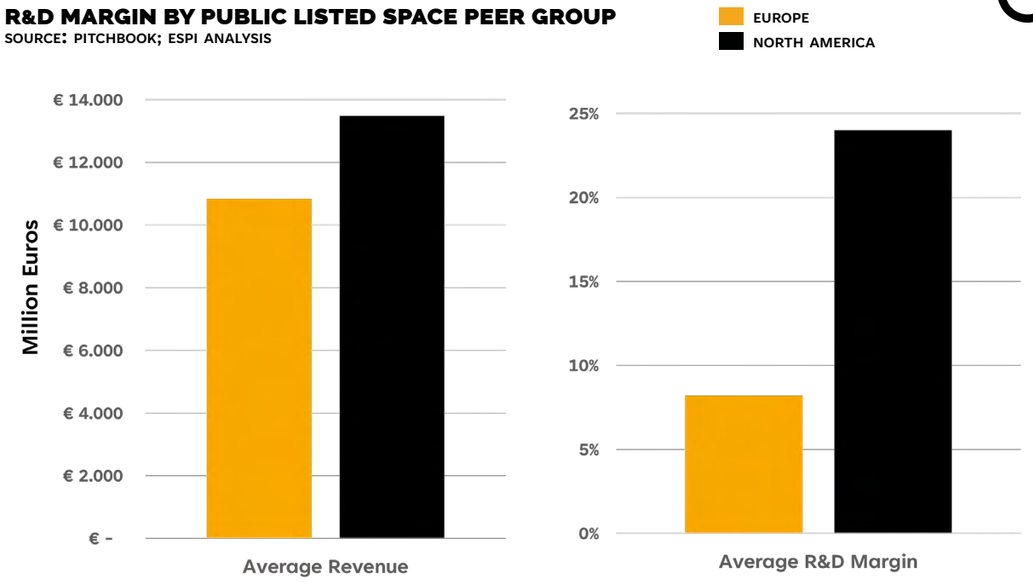
EXPENDITURE IN SPACE R&D: ADDRESSING INVESTMENT GAPS AND FRAGMENTATION

European space industry investments in R&D are not only lagging their American counterparts but are also more sensitive to fluctuations in cash flow.[8] SMEs struggle to sustain R&D efforts without financial support, especially when transitioning from product development to market readiness. Procurement contracts of space agencies frequently conflict with runaway constraints of small to mid-size players, and even artificially limit profitability levels through mandatory caps, further hindering investment in R&D.

Not just relevant to SMEs, we observe a significant gap between North American and European R&D expenditure by their respective publicly traded space constituents.[9] **Europeans spend around 1/3rd of what their North American peers spend, despite comparable revenue generation across the sector**. Average profitability differences of ~EUR 8 billion[10] between European and U.S. aerospace companies may point to a contractual reason behind this trend, with gross margins for U.S. industry higher than in Europe.

About half of European space expenditure is spent at the national level. Europe's fragmented framework based on national contracts further prevents scaling opportunities, especially considering capital markets and public grants. **By removing barriers and fostering cross-border collaboration on national contracts,** European countries could, for example, defragment public support mechanisms by making national contracts accessible to other European industrial players showing the same stage of development and offering the most competitive solution. In exchange, trade compensation mechanisms can ensure a balance of industrial benefits.

R&D MARGIN BY PUBLIC LISTED SPACE PEER GROUP
SOURCE: PITCHBOOK; ESPI ANALYSIS



The EU funding system also suffers from fragmentation, a lack of directionality, and complex governance, which hinder the ability to deliver large-scale investments impacting innovation productivity and competitiveness of European players. Europe must avoid the recurring 'phantom' direction of initiatives like the Strategic Technologies for Europe Platform (STEP), which recognises robotics, cloud computing, and smart connectivity among target investment areas, but does not meet the threshold of directed innovation policy.[11] The European Commission's identified R&D&I building blocks for space, namely: I) Use of Space on Earth, II) Access to Space, III) Action in space, IV) Monitor space, V) Explore space, and VI) Boost space, again do not provide a sufficient level of identifiable outcome to measure investment success against, nor do they provide a sufficient level of granularity to identify enabling technologies to be developed and scaled.

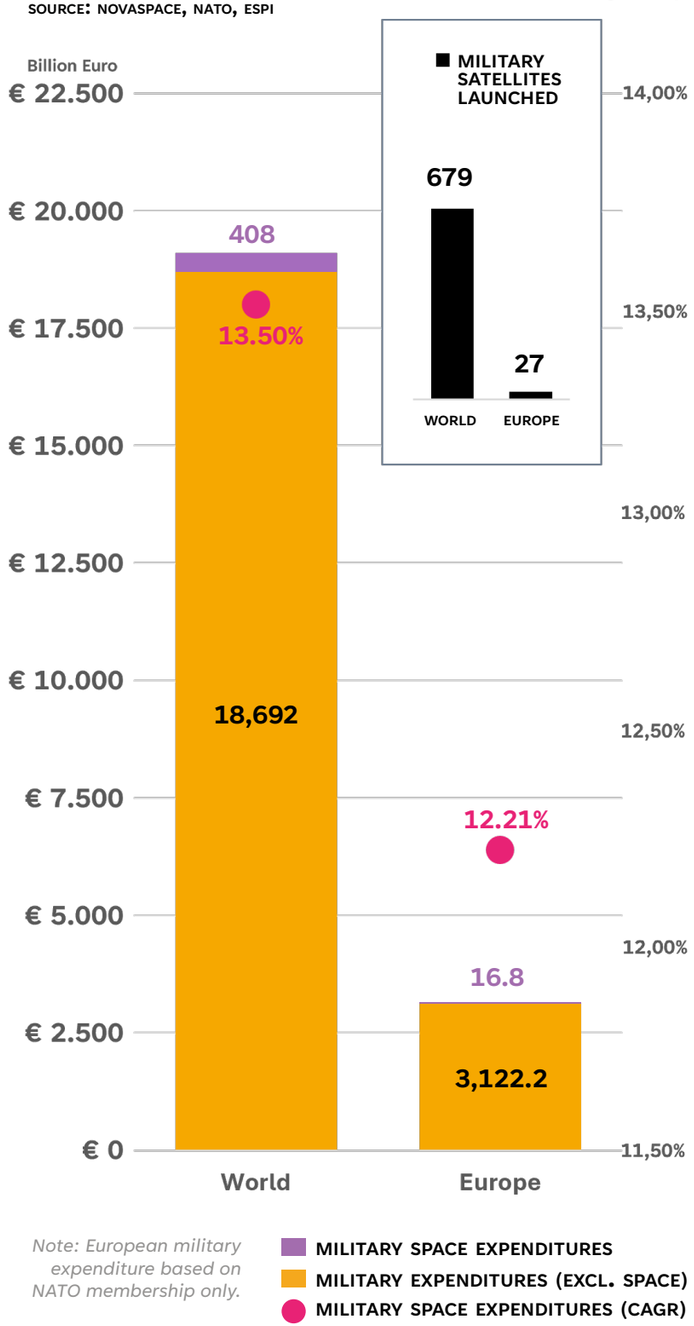
Recognising that the majority of existing instruments and schemes supporting space companies are primarily focused on pre-scale up phases of development, subsequent initiatives involving the European Investment Bank (EIB), ESA and the Directorate-General for Defence Industry and Space (DG DEFIS) must cooperate to yield deeper impact on the space sector over the next decade, in particular in scale-up phase. In this regard, the European Commission's proposal, as part of the next Multiannual Financial Framework (MFF), to streamline and simplify its innovation, is a welcome opportunity to reform towards unifying and clarifying priorities and governance across the European ecosystem.

8 *World Economic Forum, Why do Europe's young innovative companies invest so little?, May 2015*
9 *Peer group is based on public-listed space companies part of space-themed indices (JEDI, UFO, HSPCP).*
10 *JRC, The 2024 EU Industrial R&D Investment Scoreboard, December 2024*
11 *European Space Policy Institute, Bridging the Financing Gap in the European Space Sector: Alternative Funding Pathways in Tightening Markets, March 2024*

SPACE AS A SUBSTANTIAL LINE IN DEFENCE BUDGETS

Space-based technologies, including satellite communications, Earth observation and Global navigation satellite systems (GNSS) are indispensable for military intelligence, planning, and operations. The integration of space systems into modern warfare is accelerating at a rapid pace as digital technologies and Artificial Intelligence (AI) further broaden their scope of application. In addition, the shift of military operations into space poses new risks and creates needs for protection and active defence in space.

WORLD MILITARY AND SPACE EXPENDITURES (CUMULATIVE 2015 TO 2024)
SOURCE: NOVASPACE, NATO, ESPI



Note: European military expenditure based on NATO membership only.

With the multiplication of regional conflicts, military expenditures have risen significantly globally over the last 10 years, to support the development of military systems and infrastructure. While European awareness has been slow to materialise, the war in Ukraine, intensifying threats from Russia and changes in U.S. diplomatic stance have prompted European nations to take action in favour of **technological sovereignty** as a key pillar of their security. Still, Europe lags far behind both in terms of volume of investments in military space and military space assets in operation.

Let's look at who buys their products? And let's look at who they buy from? Themselves.

To strengthen Europe's defence capabilities, a substantial increase in defence spending is necessary, although spending on European military space rose by 10% between 2024 and 2023, only 15% of Europe's public space budgets are earmarked for military space activities, while the global average is about 50% and growing. This growth is also taking full advantage of increased integration of commercial solutions into military operations. At large, the **global cumulative expenditure on military space as a proportion of all military expenditure was 2.2% between 2015-2024. In Europe, that proportion was barely above 0.5%.**

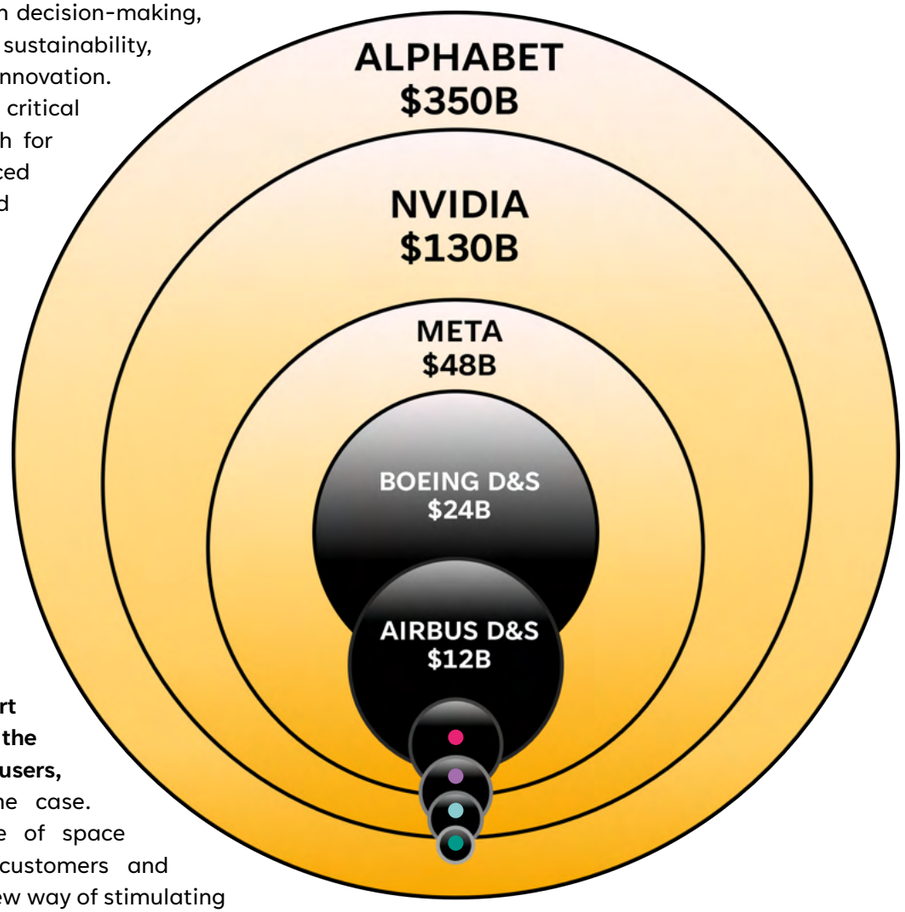
The ultra-concentration of investments in this area in Europe shows the lack of consideration and resources in most countries: the top five European countries together account for almost 90% of Europe's total military space spending, compared with about 50% for the civil space sector. It highlights that, despite the political awakening linked to security issues on the continent, there is still much to be done to translate this into concrete investment and commercial opportunity.

Strengthening synergies between the civil and military space sectors is another way to reinforce Europe's security infrastructure and economic opportunities in a sector that is, by nature, dual-use. The planned integration of defence and space as a single thematic of the new ECF operating as a single interface and rulebook is therefore a positive step that however requires caution to not overshadow best-in-class civil space programmes.

EUROPE'S SHOT AT HYPERSCALERS

To achieve global scale, Europe's space industry must overcome its market limitations and engage in collaborations. There are vast opportunities in data-driven decision-making, supply chain efficiency, sustainability, infrastructure, and innovation. Space technologies offer critical solutions as industries push for greener practices, enhanced operational efficiency, and connectivity in remote areas.

While companies have been developing B2B solutions, there is a need for increased concerted efforts and broader potential engagement with end users. This is especially true in the development of new infrastructure, such as **IRIS² or LEO-PNT, for which industries like automotive and transport should be involved early in the development cycle as test users,** which is currently not the case. Subsidising the initial use of space services for commercial customers and public authorities is not a new way of stimulating market growth. Space agencies have been doing this for decades, such as ESA through its ARTES programme, for example, but this is often done within the framework of modest projects and funding that cannot raise awareness among potential customer verticals on a large scale.



REVENUES IN 2023-2024 OF MAJOR PLAYERS IN DIGITAL AND SPACE*

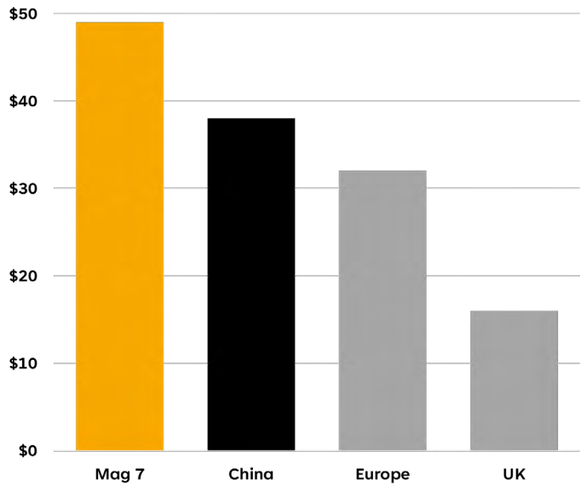
*NOT TO SCALE

To foster growth, Europe must focus on customer needs and create accessible, scalable B2B and B2C solutions. More partnerships shall be developed between space firms with other sectors, including energy, logistics, finance, biotech, agriculture, etc., leveraging corporate venture capital to overcome scaling challenges. By capitalising on its heritage and trust, Europe can expand into markets ripe for development through innovative space technologies.

> *Government budgets are important, but they are the oil in the gearbox, not the petrol that runs the engine.*

VC INVESTMENT BY MAG7 IN BILLIONS COMPARED WITH ALL VC INVESTMENT OF OTHER REGIONS IN 2024

SOURCE: DEALROOM



BLINDED BY THE LIGHT

STRATEGIC LEVERS TO RECLAIM EUROPE'S INDUSTRIAL EDGE IN SPACE

The question of skill goes well beyond the supply and demand of talent on the job market. Having access to the right skillsets and the latest technologies, applying them properly, and ensuring the long-term protection of the workforce, knowledge, and intangible assets is an increasingly strategic question.

In the space sector, where innovation, capital, and security intersect, the strength and diversity of European skills will determine Europe's position on the global stage. **This is not only about technological capabilities and know-how**, but also includes intellectual property protection, the commercial culture of public institutions, and diversity in corporate leadership.

> *The economy, is transforming at a rapid pace and our education and skilling systems cannot keep up.*
>
> *EVP Roxana Mînzatu, European Commission (2025)*

A system built around technical excellence alone will struggle to thrive in markets where changing user needs, commercial instincts, and agile decision making are equally decisive.

As global competition intensifies, building Europe's skill advantage demands a shift in how we value, develop, organise, protect, and direct knowledge across the full space ecosystem.

BUILDING THE SKILL BASE FOR A COMPETITIVE SPACE ECONOMY

Europe's strength has been historically rooted in science and engineering, but today, **technological excellence alone is insufficient to secure long-term competitiveness**. Translating research and science excellence into commercial gain is a well-known challenge in Europe.

This imbalance permeates Europe's space sector skill profile. Industry executives identify major gaps in entrepreneurship, business acumen, and market development. Financial literacy is another clear deficiency, particularly concerning innovative financing mechanisms, and strategies venturing beyond a reliance on pre-specified public programmes with a low-risk, low-reward outcome. These challenges leave the sector insufficiently prepared to adapt and thrive in a market where end customers become increasingly diverse and where integration with other sectors of the economy is essential.

> *We are still spending lots of time persuading institutional stakeholders of innovative ideas, the value of which they don't seem to be able to grasp. The more innovative we are, the less this may be possible to be squeezed into PowerPoint bullet points.*

Mobility across the public-private divide within the space sector remains low, limiting cross-fertilisation of ideas and slowing the diffusion of best practices. This challenge is particularly evident between large programme-driven institutions and commercially oriented firms, where different cultures and incentives often clash with one another. Public actors such as ESA frequently retain personnel for extended periods under indefinite contracts, which restricts exposure to external perspectives and emerging market trends.

WHICH OF THE FOLLOWING DOMAINS IN EUROPE'S SPACE SECTOR FACES THE BIGGEST SKILL GAP?

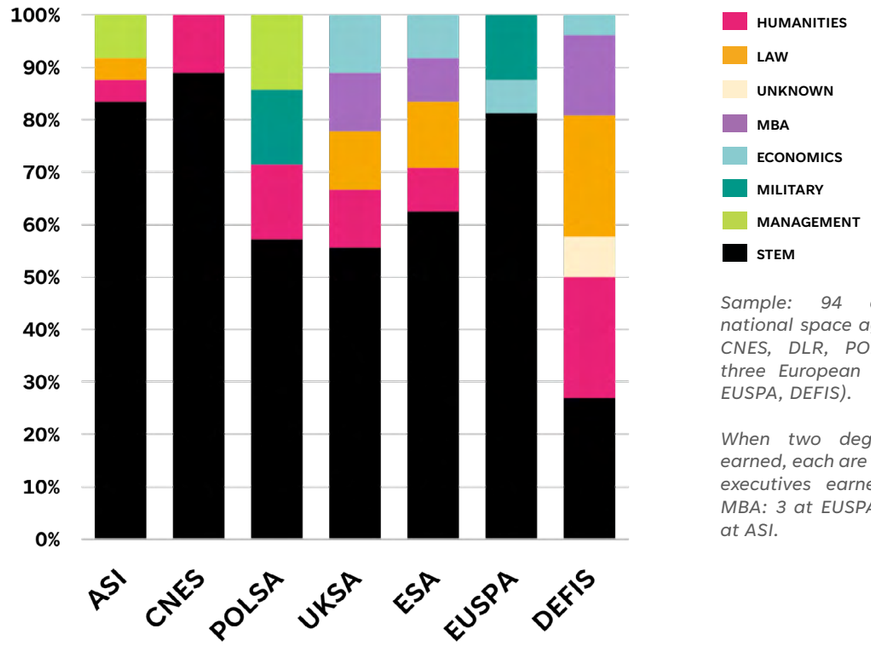
BUILDING THE SKILL BASE FOR A COMPETITIVE SPACE ECONOMY

This imbalance is also illustrated by the profiles of senior leadership within European public institutions dealing with space, where over 70% of executives have STEM backgrounds, while their **responsibilities are increasingly broad**, ranging from commercialisation, industrial policy, military interactions, and wider societal engagement.

Homogeneity can constrain institutional agility and foresight, limiting responsiveness to emerging trends and demands. To address this vulnerability, Europe must deliberately broaden its talent pool, incorporating finance experts, successful entrepreneurs and business leaders, communications specialists, and other profiles, alongside STEM professionals. Developing a broader diversity of skills, with the priority of bringing a better understanding of business dynamics and **more commercial culture into Europe's public institutions**, could alleviate some of the challenges industry faces when interacting with them. With this increased diversity, the European space sector at large would be better positioned to anticipate changes, capture new opportunities, and ensure R&D yields market-ready innovation.

EXECUTIVE LEADERSHIP BACKGROUND IN NATIONAL SPACE AGENCIES AND EUROPEAN INSTITUTIONS

SAMPLE SIZE: 94, SOURCE: ESPI



Sample: 94 executives from national space agencies (AEE, ASI, CNES, DLR, POLSA, UKSA) and three European institutions (ESA, EUSPA, DEFIS).

When two degrees have been earned, each are weighted 0.5. Five executives earned an additional MBA: 3 at EUSPA, 1 at AEE and 1 at ASI.

SHAPING CULTURE THROUGH URGENCY AND AGENCY – INSIGHTS FROM DARPA'S PROGRAM MANAGERS

DARPA's Programme Managers are drawn from industry, academia, or defence for a 2–4 years term to "seed" ideas and define, with significant autonomy, the challenges, milestones, and outcomes of their portfolio. The high-urgency/high-agency prevents ossification and combined with external recruitment pipelines, keep ideas fresh. While the focus is on innovation delivery, such a structuring of leadership intake could be tuned to bolster a commercial culture.

Taking ESA as an example, limited-tenure roles for "Commercial Programme Managers" should consider a maximum possible impact within long-duration programme management. Less about an end-to-end ownership, the role could therefore focus on the commercial architecture of a programme and its early execution phases.

Such a dynamic framework of industry secondments, already considered in the early days of the European Space Agency, could be plugged into existing structures, both within ESA, the European Commission and national agencies.

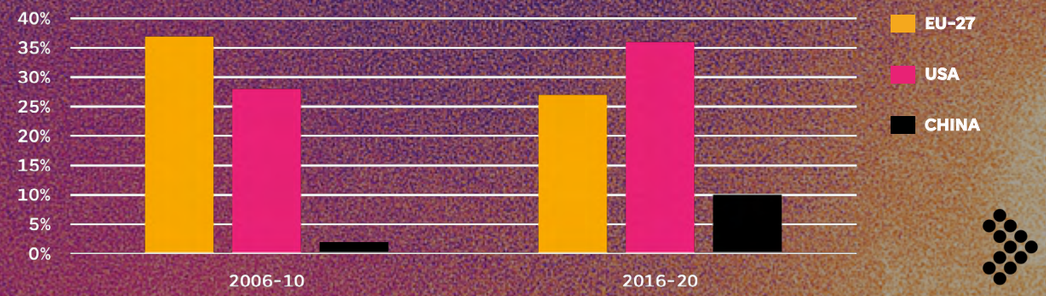
PROTECTING STRATEGIC KNOWLEDGE ASSETS

Protection of Intellectual Property Rights (IPR) is essential to turn innovation into successful commercial products. While filings do not guarantee market success. Patent activity is nevertheless a useful proxy for determining the innovation potential.

Over the past two decades, Europe has fallen behind in patent filings, while global activity in space technologies has grown. Areas where Europe once led, showcased by filings across the five major patent

IP-5 PATENT FAMILIES FOR SPACE-RELATED TECHNOLOGIES

SOURCE: OECD



PROTECTING STRATEGIC KNOWLEDGE ASSETS

offices (IP5 patents: Europe, Japan, Korea, United States, China), both in general space-related technologies, and specifically in propulsion, have seen its advantage erode. Europe's research base remains strong, but the ecosystem needs to find a new footing to be a global leader.[12]

Europe's slower pace may reflect both declining competitiveness and a weak commercialisation strategy. When research has clear economic value, organisations are more likely to file patent applications to protect it. In this context, the **ESA-industry IPR model is a strong template for public-private collaborations**, granting the industrial partner ownership of IP generated under ESA contracts, aligning incentives to further invest and commercialise the developed products and services.

Similarly, Europe risks being wrong-footed in the future satellite spectrum landscape. Beyond X-band, highly valued for military and government applications, spectrum is not fully considered as a strategic asset in its own right. Other bands, such as L-band and S-band, are extremely important for mass market D2D services. Still, with limited exceptions, it does not seem to be considered crucial for Europe's long-term competitiveness. One example of this under-acknowledged role is the May 2024 BEREC external workshop on

AN ERODING LEADERSHIP IN PATENT FILINGS

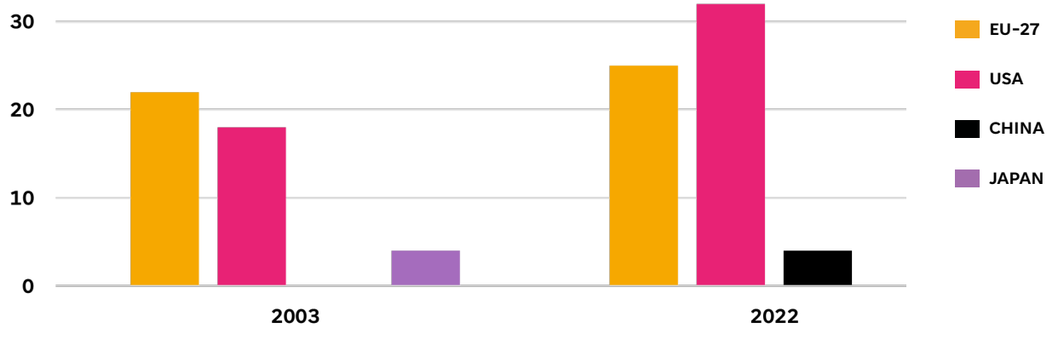
PROPULSION
Propulsion technologies are a field where Europe has traditionally held a leading position thanks to its heritage of proven designs and advanced manufacturing capabilities. This is evidenced by filing the most international applications in 2003. However, this was eroded over the years, and by 2022, the U.S had already replaced Europe as the number one global international patent filer.

QUANTUM
The U.S. and China (most filings being national) lead the way, with over 300 rights for each in the 2001-2020 window. Despite strong technological capabilities and significant investments through initiatives like EuroQCI, Europe comes in third, with fewer than half the filings. This exposes a weakness and poses a question on Europe's ability to keep a commercial edge in critical emerging technologies.

SPACE-BORNE SENSING
During the 2010s Europe saw its international applications in remote sensing technologies stagnate. Over the same period, the rest of the world increased the number of annual patent applications by a factor of two. As a result, by 2021, whereas the U.S. maintained its global leadership, Europe had already been surpassed by China.

Source: Joint reports by the European Patent Office, the European Space Agency and ESPI (2021 – 2024)

INTERNATIONAL PATENT FAMILIES BY APPLICANT COUNTRY IN PROPULSION SYSTEMS
SOURCE: EPO, ESPI



the usage of satellite technologies in mobile communications. It featured presentations from 6 U.S. satellite operators, out of 8 U.S. representatives in total, while **not one European satellite operator presented their positions** or future plans.

In just under 2 years, European licenses for the 2 GHz spectrum (S-band) for mobile satellite service will expire, and the question of renewal or reallocation is open. Today, this spectrum is held by U.S. operators Viasat and EchoStar, following their acquisition of Inmarsat and Solaris Mobile, respectively. With the competition for spectrum targeting D2D markets heating up, exemplified by the $17 billion acquisition of spectrum licenses by SpaceX from EchoStar, 2027 can be a decisive year for Europe's long-term competitiveness in satellite connectivity.

PARTICIPANTS TO THE 2024 BEREC WORKSHOP ON SATELLITE TECH IN MOBILE COMMUNICATIONS
NO EUROPEAN SATELLITE OPERATORS

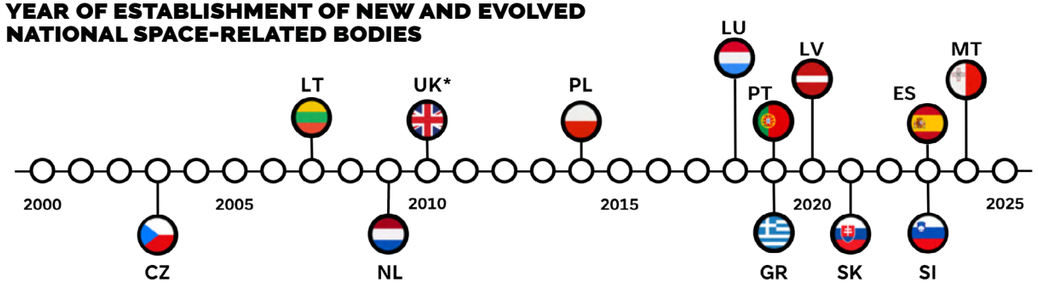
12 *OECD, The Space Economy in Figures – Responding to Global Challenges, September 2024; European Patent Office, European Space Policy Institute, ESA, Propulsion Systems for Space – Patent insight report, May 2024*

EUROPE'S SHIFTING CENTRE OF GRAVITY OPENS DOOR FOR NEW DYNAMISM

Over the past 20 years, the European space industry has changed significantly, expanding and deepening its capabilities across the continent. **New regional hubs of knowledge and capabilities are emerging**. If provided with the necessary resources and capacity to realise their ambitions, Europe as a whole could benefit from this new pool of skills, with the potential to stimulate new dynamism in the space industry.

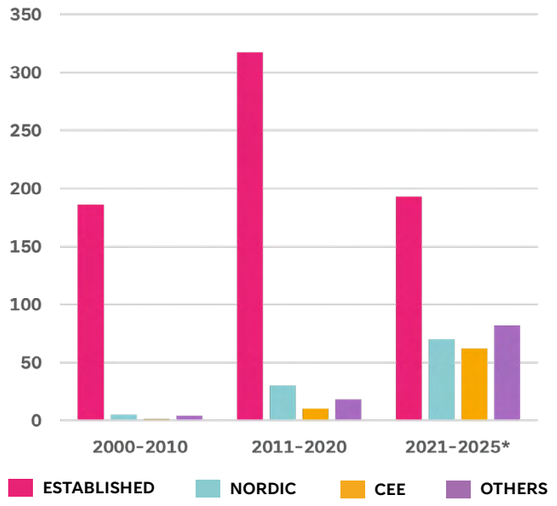
At the institutional level, this geographical distribution of capabilities is evidenced by the creation of a growing number of new or evolved national institutions dealing with space across Europe, signalling a growing ambition of an increasing number of European countries to develop their space capabilities and reap the associated socio-economic benefits.

YEAR OF ESTABLISHMENT OF NEW AND EVOLVED NATIONAL SPACE-RELATED BODIES



The geographic spread of industrial capabilities is evident in the number of satellites launched, when classified by the country of origin of the manufacturer, pointing to a clear **dispersion of the ability to build complete space systems**. Whereas in the 2000-2010 period virtually all systems were produced by established countries (France, UK, Italy, Germany), in the 2010s, this started to change as small satellites and new production models diversified manufacturing capabilities. Over the last 5 years, satellites produced by European manufacturers became more evenly distributed across countries, including the Nordics and Central and Eastern Europe. For the first time, manufacturers in emerging space ecosystems built more satellites than in established hubs.

LAUNCHED SATELLITES BASED ON THE ORIGIN OF SATELLITE MANUFACTURER



One of the most notable changes at the industry level has been the level of know-how and ability to master the full value chain, thus **gaining direct access to institutional customers for full space systems**. This confers greater capacity to emerging companies in non-traditional space hubs, establishing their vision for technologies, products and services of the future.

Large System Integrators (LSIs), such as Airbus D&S, Thales Alenia Space or OHB, located mainly in France, Italy, the UK, Germany, Spain, Belgium and the Netherlands, continue to have a structural role. Nevertheless, many companies in non-traditional geographies have the ambition to not only be a part of the supply chain but to manufacture and deliver complete systems.

Upcoming investments in defence, which will trickle down to the space sector, will likely drive these dynamics even further. Defence investment will increase, especially in countries that are close to Europe's security threats, not necessarily the same ones that have been traditionally seen as the main hubs for space development. Moreover, fiscal room to invest will also play an important role in developing future space clusters in Europe. In this scenario, countries such as those in the Nordics and on Europe's Eastern borders are expected to continue shifting the centre of gravity in the European space sector.

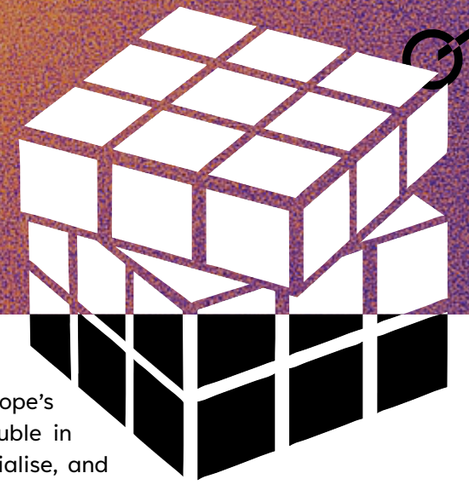
EXAMPLES OF COMPANIES CAPABLE OF DELIVERING SPACE SYSTEMS IN NON-TRADITIONAL INDUSTRIAL CENTRES

- ICEYE
- Creotech
- EnduroSat
- GomSpace
- NanoAvionics
- Skylabs

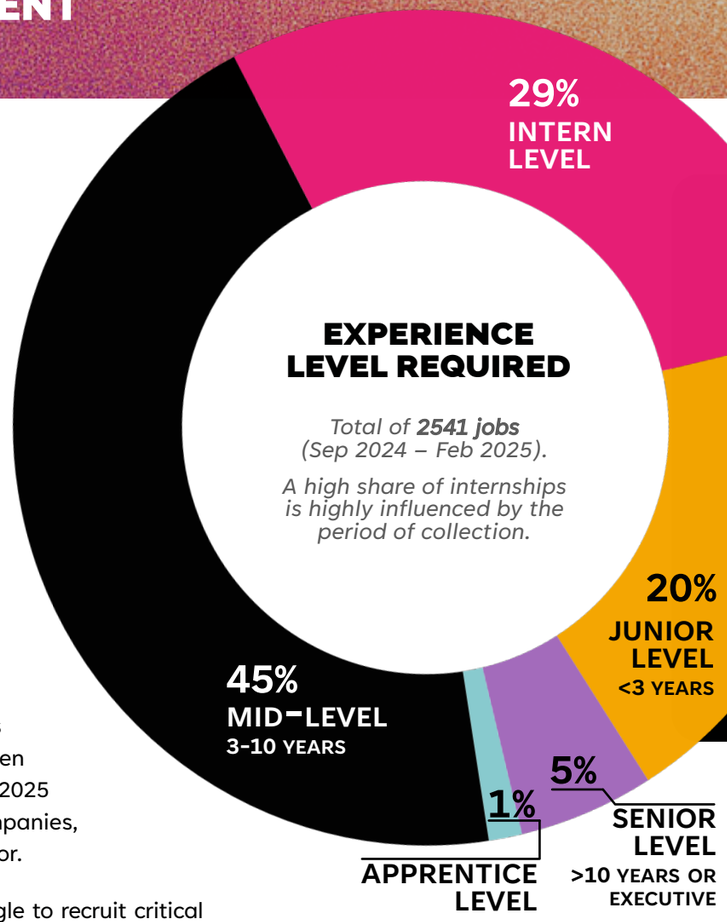
IN WHICH PART OF EUROPE DO YOU EXPECT STRONGEST SPACE INDUSTRY GROWTH OVER THE NEXT 10 YEARS?

ESPI INDUSTRY WORKSHOP PARTICIPANT VIEWS ON EUROPE'S SPACE INDUSTRY GROWTH IN RELATIVE TERMS

AMIDST A GLOBAL COMPETITION FOR STEM TALENT



Parts of the European space sector are already under labour market pressure. Between 2021 and 2024, the upstream industry added more than 13,000 Full-Time Equivalent (FTEs), driven by the rise of space start-ups and increased public investment.[13] This growth has increased the pressure on hiring skilled workforce – especially for mid-level profiles. Mid-level positions represent almost half of the almost 2500 vacancies analysed by ESPI between September 2024 and February 2025 across a (limited) set of 33 companies, one Agency, and one aggregator.



> *Europe faces a growing challenge in meeting the demand for skilled talent in STEM fields, especially in strategic sectors like clean and circular technologies, digital technologies, aerospace and defence*
>
> *The Union of Skills, European Commission (2025)*

Larger firms increasingly struggle to recruit critical technical profiles particularly for software-related roles and Radio Frequency (RF) engineers. This is exacerbated by an overall slump in STEM graduates between 2015 and 2022, with a decrease of -7% in doctoral graduates in STEM fields, as well as a general decrease in natural sciences, mathematics and statistics (-13.1%), information, and communication technologies (ICT) (-25.5%), while increasing in engineering, manufacturing and construction (+9.4%), according to the European Commission.[14]

13 *ASD-Eurospace, Facts and Figures 2024, April 2025*
14 *European Commission, The Union of Skills, March 2025*

If estimates that Europe's space budget could double in the coming years materialise, and with a generational retirement wave looming across agencies, the shortage on the labour market will only intensify.

With more programmes to implement, more complexity to manage, a larger workforce will be needed to do it. Companies will need to pursue with new hiring models, training programmes, and attractive career pathways to retain and scale their workforce, especially in segments where the space sector is competing with higher-paying high-growth industries. To do so, space must use the **full power of its inspirational dimension**, and **attract professionals from other sectors**, bringing new dynamism, knowledge and preventing a closed circle effect.

Companies will need to get creative, experimenting with new hiring models, training programmes, and new career pathways to retain and scale their workforce. Without this, labour constraints risk becoming one of the biggest bottlenecks to Europe's space ambitions.

BUILDING A TAILORED PIPELINE OF NEXT-GENERATION TALENT

EnduroSat's *Space Challenges Bootcamp offers an industry-led, hands-on 1-month training programme designed to fast-track junior profiles into operational space roles. With a strong focus on systems engineering, software, and mission operations, it bridges the gap between academic education and industrial requirements. Not only does the programme align with the company's mission in Education, it also supplies EnduroSat with a pipeline of pre-vetted talent, with the Bootcamp yielding approximately 30% of the company's current workforce.*

BLINDED BY THE LIGHT

STRATEGIC LEVERS TO RECLAIM EUROPE'S INDUSTRIAL EDGE IN SPACE

Vision is where strategic foresight meets policy direction and programmatic alignment, not just defining what Europe wants at a declaratory level but defining how it plans to get there.

> *Governments cannot solve today's challenges on their own. Industry and business must also jointly contribute to policy design and ensure effective implementation in private sector development.*
>
> *UNIDO, Industrial Development Report 2024*

As Europe is shaped by national, intergovernmental, and supranational actors, political will is indispensable. But political alignment alone is not enough. Europe also needs a determined and globally driven industrial ambition, seeking to expand and challenge today's market leaders across the full spectrum of space products and services.

A successful vision needs a broad buy-in from both institutional and industry actors. This requires more than strategies and roadmaps. It needs a clear purpose, mechanisms for collective agenda setting, a smart Europe-first regulatory framework, and the bravery to make hard trade-offs.

REBALANCING THE ROLE OF INDUSTRY IN SHAPING EUROPE'S SPACE AGENDA

The active participation of industry is essential to developing a forward-looking strategic vision for the European space sector. However, this has been **severely lacking to date**, leading to missed opportunities to collectively contribute to long-term capability or programme scenarios. The space industry should be a central partner of government agencies in shaping the strategic vision for the European space sector, rather than a passive or peripheral participant.

Established companies and start-ups must be encouraged to engage actively and share their entrepreneurial vision based on their strengths and resources.

A cohesive Europe-wide industrial policy for space is crucial, especially considering the multi-layered, institutional landscape of the European space programme. The industry's central role is particularly important given the growing influence of the defence sector on space industry dynamics. **From its heritage to serve both civil and military solutions, the space industry is uniquely positioned to serve as a bridge between space agencies and the defence industry, facilitating a more integrated and strategic approach.**

> **A cohesive Europe-wide industrial policy for space is crucial, especially considering the multi-layered, institutional landscape of the European space programme.**

Currently, there is no proper structured and continuous forum that allows the European space industry to contribute to the development of a long-term vision for the space sector. Industry is consulted, invited to participate in surveys, and talks at conferences, but discussions do not take place within a structured framework, which prevents it from contributing effectively to policymaking.

Existing forums are often fragmented, inefficient, and self-restricting to individual space domains or applications, highlighting the need for a more integrated approach. There is a pressing need to establish a framework for continuous dialogue involving the industry, the EU, national policymakers, ESA and national agencies. While some countries designed structures for such dialogue (e.g. US National Space Council Users' Advisory Group), a similar, cohesive structure is required at the European level.

SAFEGUARDING STRATEGIC ASSETS IN A VOLATILE GEOPOLITICAL AND FINANCIAL LANDSCAPE

ADVANCES IN INNOVATION FAIL TO TRANSLATE INTO EUROPEAN COMMERCIAL GAIN

In November 2024, with support from the Communications, Space & Technology Commission (CST), U.S. incorporated Viasat tested direct-to-device (D2D) satellite messaging in Saudi Arabia, demonstrating how D2D can fill gaps in remote areas, a projected $50 billion market by 2032. Two months later, at the European Space Conference, Viasat and ESA agreed to design an open-architecture LEO network for 5G Non-Terrestrial Networks (NTN) services, co-funded under ARTES. Building on this, Viasat and the UAE operator Space42, announced plans in March 2025 for shared, multi-orbit 5G NTN infrastructure to speed up deployment, cut costs, and integrate with terrestrial operators, framing the agreement as a direct building block of the ESA partnership. In September 2025 the companies announced Equatys, a joint venture to enable global D2D services.

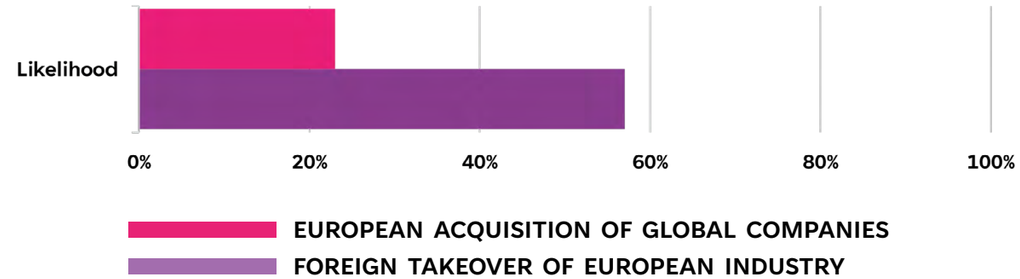
Europe's citizens, although funding the R&D grant, might see little direct benefit from the commercialisation of D2D, while European user industries might need to exclusively depend on foreign service providers.

ESA's funding under ARTES advances the satellite integration goals and successfully shapes standards, but deployment and commercial gain are happening abroad. Europe risks remaining a secondary player, exporting innovation while being left out of the first-mover advantage it funds. ***Without reformed thinking Europe will continue to incubate technology for foreign markets.***

European space industry players feel more like prey than hunters in a consolidated global market environment. While Europe possesses world-class technological strengths, particularly within innovative SMEs and midcaps that excel in cutting-edge sectors, paradoxically, these strengths also leave Europe exposed. As many companies are rich in technology but poor in capital, they become attractive targets for foreign acquisitions, often losing their independence, strategic assets (IPR, know-how, spectrum), and growth prospects. In a market where technology alone is not enough,

Europe's vulnerabilities become acute. In the absence of strong domestic demand and regulatory safeguards, high-tech innovation can, in the long-run, turn into a new strategic dependency.

A survey of industry leaders suggests a widespread perception that European firms are at constant risk of acquisition due to chronic capital shortages, limited long-term funding security, and fragmented demand. Addressing this perception requires the **development of new mechanisms; more integrated FDI screening for the space sector, dedicated investment vehicles, protective policies, or incentives that reward long-term capital allocation to strategic sectors**. Only by doing so can Europe sufficiently safeguard its strategic assets and position its industry not just to survive but actively shape global market dynamics.



Several countries demonstrate how strategic technology sectors can be shielded from premature foreign exposure. By temporarily strategically restricting foreign access until domestic industries achieve critical maturity and resilience, markets can safeguard and strengthen their capabilities before engaging in global competition. Europe can draw from this playbook with measures that carefully sequence openness and protection; yet such an approach requires growing and strategically deployed European demand.

RECLAIMING SPACE AS A TOOL FOR PEACE, PROSPERITY AND GLOBAL RELEVANCE

Europeans are obsessed with the U.S. and concerned about China. To the contrary, Europe is largely absent from American and Chinese considerations when it comes to formulating their national policies and objectives, including in the space sector as in others. The American AI Act published in July 2025 includes a full chapter dedicated to China, while it has no single reference to Europe. Europeans frequently use the U.S. as a benchmark for competitiveness, but it is high time for Europe to define a vision that looks beyond threat-driven competition with the U.S. and China and assumes responsibility for European leadership.

The European space sector is currently navigating a period marked by rapid technological advancements and sectoral transitions, posing unique challenges to its traditional model. The U.S. and Chinese strategies are fuelled by **a vision of technological dominance, viewing space, like other disruptive technologies such as AI, as tools for global power.** As emerging commercial solutions push technology to new frontiers previously only pursued by governments, national security finds itself increasingly bound to private actors, their innovation potential and industrial vision.

> *Today, the entirety of economic and military space activities is confined to the geocentric regime; however, commercial investments and new technologies have the potential to expand the reach of vital National space interests to the cislunar regime and beyond in the near future. As technology marches forward, U.S. military spacepower must harmonize with the other instruments of power to protect, defend, and maintain the Nation's strategic interests in space.*
>
> *Space Capstone Publication, US Space Force (2020)*

In order to maximise this relationship, governments are aggressively moving forward to relax environmental, societal and regulatory standards, often overlooking potential negative externalities.

Europe must strive to lead in productivity growth and innovation of its space sector, guided by its unique model and core values. An approach that includes prosperity, social inclusion, freedom, peace, and democracy within an open economy and a sustainable environment, will yield better long-term results by avoiding pitfalls brought about instability and erratic decision-making. It is time for Europe to propose an alternative vision, strong and proud of its values.

Without compromise.

A SYSTEM THAT FUELS ITSELF – AN ECOSYSTEM APPROACH TO EUROPE'S SPACE SECTOR

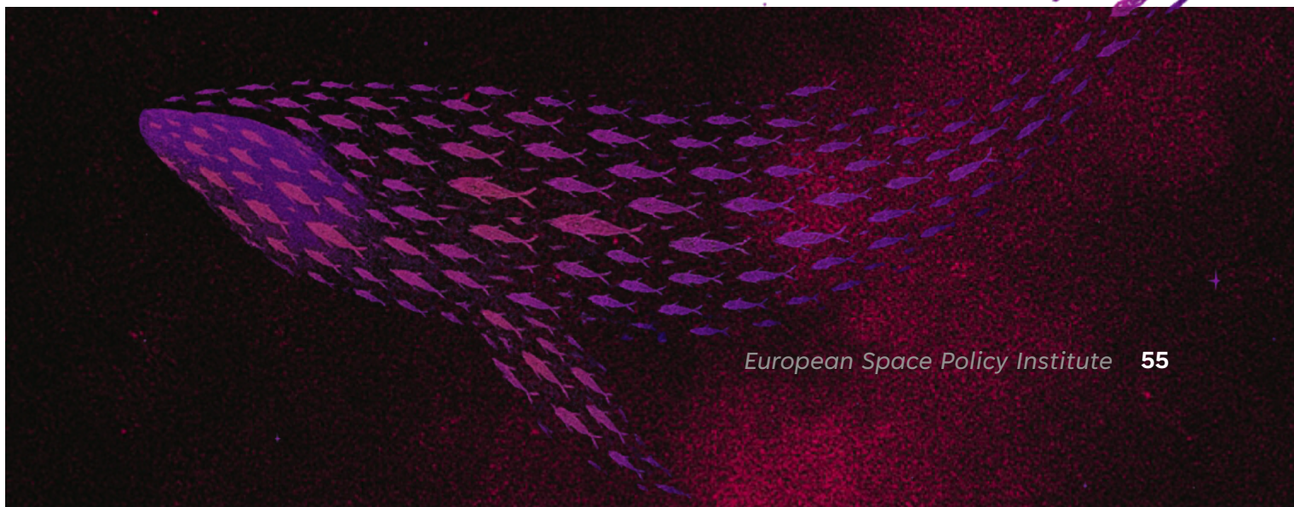
A forward-looking European space vision must move beyond individual flagships and isolated technologies. A vision is much more than a space programme. It is a force that leverages and catalyses a much wider ecosystem than the confined one of the space sector.

Europe's space landscape remains siloed, with links between policymaking, R&D, education, finance, and industrial actors developed ad-hoc and lacking an integrated vision. In Europe, groundbreaking innovations too often **stall between the laboratory and the marketplace**. And even when they do reach commercial success, they rarely attain broader adoption.

> *It takes a village.*

Europe's vision for space must therefore cultivate an integrated ecosystem approach that emphasises not just what Europe intends to build and project, but how it will engage and empower the full range of non-space industry players, academia, research centres, government institutions, investors and citizens. A vision must ensure each breakthrough stimulates further innovation, attracts fresh investment, nurtures new talent, and fuels greater ambition – an ecosystem of compounding growth and positive energy that can inspire Europeans and instil a sense of achievement and pride across the continent.

Such an approach demands coordinated policy actions involving top European political leadership, including Heads of State and influential thought leaders. But it also requires **industrial leadership to step from the shadows, openly advocate for bold ideas, and drive the European space ecosystem beyond the confines of immediate shareholder interests and quarterly results**. Short-term maximisation of shareholder value works only for so long before corporate decline inevitably sets in.



FROM BOARDROOMS TO BREAKTHROUGHS?

ESPI European Space Policy Institute

During 2025, major European stakeholders have taken steps or are in advanced discussions towards a consolidation of the European industrial space ecosystem. This is part of an overdue response at the highest level of industry and governments, as industrial and geo-political competition is already out-performing European capabilities. In a first instance, it is often focused on responding by scaling-up, mostly through sectoral, horizontal integration, predominantly in manufacturing and in satellite operations.

Consolidation is a normal process in many industries, and scale can be expected to improve cost efficiency and economies of scale. Space is no exception. In addition, in times of political support for increased European autonomy and the required industrial capabilities, the consolidation of the space sector finds governmental backing, which in turn can leverage additional private funding.

This can be observed by EUTELSAT's recent capital increase, to which the French state contributed roughly half, and the €1B 10-year contract with the French military. At about the same time, the acquisition of INTELSAT by SES was concluded, combining two of the largest satellite operators into a single entity with a combined revenue just shy of €4B. Ongoing discussions at the highest level on project BROMO, the potential merger between Airbus D&S, Thales Alenia Space & Leonardo's space division, complements the picture.

The findings of this report suggest that these first, important and welcome steps and discussions on European industrial reform need to be followed up and complemented by addressing further root causes of the disruptions, which triggered this consolidation in the first place, beyond the aspect of scale and cost efficiency.

This is furthermore supported by looking at earlier corporate tie-ups in the U.S., which focused on scale alone and have not created market leaders, but rather companies that are today themselves disrupted by SpaceX. More of the same did not deliver.

Besides mere scale, global disruptors at their core are characterised by vertical integration of the full space value chain from launchers to infrastructures, to services – e.g., SpaceX's F9 and Starlink (and Tesla) – and the integration of space into the digital economy, e.g. Blue Origin, Amazon Kuiper, and eCommerce.

In Europe, further steps to address the above are needed and expected by the market, as signalled by the limited reaction of financial markets to the measures brought forward so far, incomparable with the steep hike in stock markets in March 2025, following European political concerns regarding the availability of Starlink services in Ukraine. Furthermore, the news that the UK was contemplating a capital increase in an operator saw the stock price practically static, despite strong political signalling.

The presently rather limited effect on market actors also stems from open questions about the effects on parts of the space business beyond satellite manufacturing, like exploration and geointelligence, on commercial markets beyond governments, and on new entrants. Notably and no less important, questions also remain open on possible negative side-effects on the speed of innovation and agility of action, key to entering new markets.

European continued efforts in consolidation need to provide a balance between efforts often perceived as focused on scale and CAPEX and OPEX-related synergies, and efforts to ensure agility and vision of all actors for capturing potential future mass markets, which are already manifesting themselves with billion-dollar investments by Apple, joint ventures between AST SpaceMobile and Vodafone, Space42 and Viasat, and SpaceX's recent $17B spectrum deal with EchoStar.

Scale and horizontal integration of segments of the European space ecosystem might be a necessary but non-sufficient step for Europe to regain its industrial edge and be part of the next cycle of innovation in space.

THIS REPRESENTS ESPI'S OWN INDEPENDENT TAKE AND DID NOT FEATURE IN THE TWO INDUSTRY WORKSHOPS.

SHIFTING INDUSTRIAL LANDSCAPES IN EUROPE

SPOTLIGHT: GERMANY

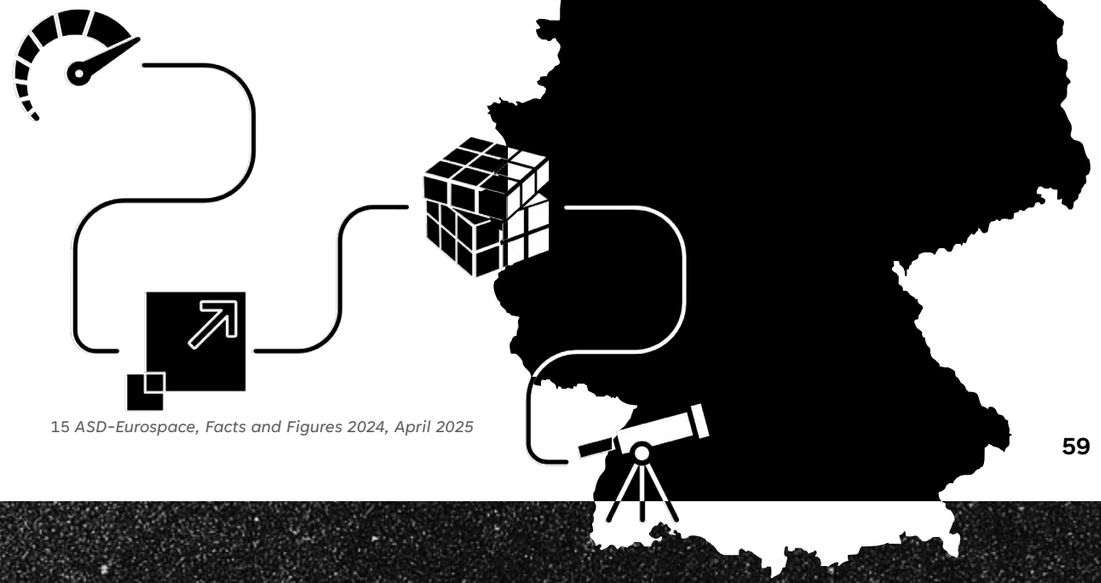
As this study is being issued, the announcement by the German Minister of Defence to invest €35B into space by 2030, at par with the full annual budget of ESA, represents a major new development, which will impact the topics addressed in the report. The size of investment and short timescale can be expected to result in major changes in the political and industrial landscape, defining space in Europe beyond the current decade.

A few initial reflections are provided below, while further work will be required to fully understand the impact, once more information becomes available:

- The initiative provides a unique opportunity to both (i) strengthen European and global **partnerships**, especially in support of increased resilience from space for critical infrastructure protection, and against hybrid attacks, as well as (ii) to develop dual-use synergies for commercial solutions for European and export markets.
- Germany's **industrial capacity** would be expected to double or more, across the entire value chain from launch to services, and impacting startups, SMEs, and established enterprises, with the defence industry in the starting blocks, ready to engage as explored in ESPI's Report on the (re) convergence of space & defence industries.[15] An open question remains if/how Germany would address its lack of a national satellite operator serving security as well as global commercial markets.
- **Workforce** and talent development will be a real challenge as a sizeable increase of human resources in 2–5 years will be required, with a high percentage of talent to be attracted from **non-space sectors** (e.g., AI, defence, automotive).
- To ensure **delivery on time and scale** and secure access to critical raw materials, technology, and the full spectrum of industrial solutions (beyond German leading capabilities in civil Earth Observation), the industrial ecosystem will need to secure **resilience of supply** and also procure **European solutions as a first priority** where national industry falls short.
- Industrial solutions should allow **the interoperability or joint use** with EU-level activities and new space programmes of countries like Poland or the Nordics, including **different degrees of command & control.**

- Complementary to large-scale procurements expected to dominate the announced investment, **industry-driven innovation** and **business agility** will need to be ensured to secure a sustainable industrial base in Germany and Europe, vital for mid-term competitiveness, beyond 2030 when public support may thin out.
- To ensure on-time delivery and performance, **Institutional support** by agencies like DLR and ESA, including for in-orbit demonstrations and the development of minimum viable products, should support industrial innovation and validation for next generation solutions.

Germany's large-scale investment has the potential to **boost and positively reshape Europe's space ecosystem.** It represents a unique opportunity to build a sustainable, competitive, European industrial and institutional fabric for space that extends beyond 2030.



15 *ASD-Eurospace, Facts and Figures 2024, April 2025*

RECOMMENDATIONS

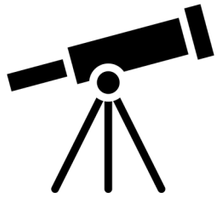
In view of the analysis provided in this report and associated workshop discussions, a set of recommendations, primarily targeting European institutions and decision-makers in line with ESPI's mandated purpose, is proposed below. The recommendations span procurement, innovation, economic security, talent, and future demand, emphasising the joint responsibility of public and private stakeholders for Europe's future.

FAST TRACK PROCUREMENT

Apply the lessons of the U.S. SDA's PWSA by integrating time-bound and high-pace development cycles (e.g. ≤18 months to prototype delivery) into Europe's civil and military space development and procurement programmes. As a result of anticipated demand, this would ensure a continued pipeline of innovation, faster deployment and a stronger and more empowered industrial base, stimulating competition across a diverse set of lead actors and supply chains and allowing agility to respond to evolving needs.

BUYING EUROPEAN

As practised in different shades across major space powers globally, establish a clear distinction of European vs. Europe-based procurement, prioritising industrial bids and service offers of companies headquartered in Europe, and under European control. This enables scale and long-term demand, and more importantly, ensures long-term benefits of publicly backed innovation, with a positive impact on Europe's competitiveness.

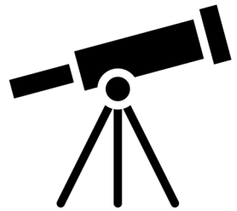


SCALING INNOVATION

The space sector needs to ensure and scale a broad and dynamic innovation ecosystem that exists separately from the main procurement-driven programmes. In a global race for innovation, the growth of such schemes needs to (at minimum) be in proportion of Europe's aggregate space budgets, recognising that early-to-mid TRL activities are inherently dual-use and underpin both Europe's civil and defence ambitions in space. The ESA GSTP programme is one such example that should be expanded and paired with direct fast-track links to larger programmes such as Earth Observation, Navigation, and Secure Connectivity.

MOONSHOTS

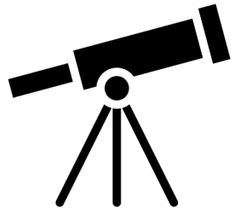
Beyond innovation applicable in the next 5 years, Europe also needs to dare to invest in long-shot technologies, applications that might be 10 years away but might bring about the next wave of transformative change. The inclusion of the space economy as a candidate in the 2028-2034 Horizon Europe Moonshot portfolio needs to be translated into bold industry-led proposals for game-changing applications capable of positioning Europe as a global pioneer.



RECOMMENDATIONS

ECONOMIC SECURITY

It is vital that Europe does not lose its key strategic assets to foreign buyouts and influence. A European Strategic Space Fund should be established to act as a last resort to step in as a temporary caretaker when high-risk foreign acquisitions are on the cards, mandated to find a strategically better ownership fit.



FUTURE OF TALENT

When defining new European programmes and flagships, these need to include ring-fenced budget lines devoted to the hiring and development of new talent and partnerships and employee exchanges between space agencies and the private sector, to ensure long-term programmatic continuity. This should be complemented by strong upskilling and reskilling initiatives from lateral sectors and a boost to EU Blue Card processes with faster procedural deadlines and employee safeguards.

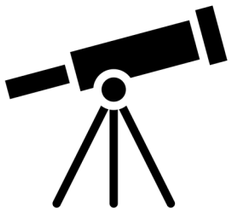
The recommendations have been developed independently by ESPI, in the belief that their implementation will contribute to a better mix of speed, scale, skill and vision, enabling Europe's space sector to continue its transformation, reclaim its competitiveness, and once again set global standards of innovation.

FROM PUSH TO PULL

To expand the use of space applications across industry and policy domains such as automotive, transportation, pharmaceuticals, defence, and agriculture, public procurement should provide incentives for the space industry to collaborate with non-space sectors. This can include preferential treatment for mixed consortia or full-stack solutions (e.g. for multi-Domain Operations in defence), or through the provision of space-service subscription support for a limited time period. The efforts can be complemented by a Satellite Users Association fostering exchange between satellite users, service providers, equipment manufacturers, and operators and doubling down on the customer-focus paradigm.

A SHARED VISION, A SHARED RESPON-SIBILITY

Create an open platform supporting a dialogue between industry and institutions to help co-shape Europe's long-term space vision and industrial action. This platform would provide the wider industry spectrum with a structured voice for better alignment between political goals, market realities, and strategic autonomy needs. It could regularly provide its views at suitable occasions, industry-driven events like ESA Industry Days, the European Space Conference, Weltraumkongress, or CASSINI Entrepreneurship Days. The forum could continuously monitor a set of high-level KPIs on Europe's performance across civil, commercial and military space ambitions, and would be managed by an independent secretariat.



IT TAKES A VILLAGE

The report was informed by two ESPI workshops, held in Luxembourg (May 2025), and Berlin (June 2025) respectively. The workshops brought together ESPI Members, ESPI Advisory Council representatives, and a host of active and former industry executives representing the full value chain and reflecting a broad geographic distribution.

DAMON OLLOMON
ICEYE

DENNIS MOORE
REFLEX AEROSPACE

IZTOK KRAMBERGER
SKYLABS

ANJA N. PECUJLIC
ENDUROSAT

MIGUEL A. GARCIA PRIMO
HISDESAT

MATTIA MOSCARDINO
GMV

ELIZABETH SEWARD
EX-BAE SYSTEMS

JOSÉ SÁNCHEZ RUIZ
HISPASAT

TOM SEGERT
BST

GRZEGORZ BRONA
CREOTECH

JAN-MARC NASR
EX-ADS

MACIEJ KRZYZANOWSKI
CLOUDFERRO

LINA HOLLENDER
CONSTELLR

GIOVANNI PANDOLFI
LEAF SPACE

JULIEN LAMAMY
ISPACE EUROPE

ESPI CONTRIBUTORS: Hermann Ludwig Moeller, Matija Rencelj, João Falcão Serra, Jermaine Gutierrez, Julia Drössler, Bianca Santi, Valentin Hoffmann, Clelia Iacomino, Anastasiia Posnova

GIANALDO MANTOVANI
OHB SE

SERGIO GRECO
THALES ALENIA SPACE

MATHIAS LINK
LSA

FRITZ MERKLE
ESPI ADVISORY COUNCIL

MAGALI VAISSIÈRE
ESPI ADVISORY COUNCIL

SINÉAD O'SULLIVAN
ESPI ADVISORY COUNCIL

NICOLAS CAPET
ANYWAVES

EVERT DUDOK
EX-ADS

GIOVANNI SYLOS LABINI
PLANETEK ITALIA

ADEL HADDOUD
INFINITE ORBITS

PEDRO RODRIGUES
TEKEVER

JUAN TOMAS HERNANI
STALATIS

OMAR QAISE
OQ TECHNOLOGY

ANIBAL VILLALBA
PLD SPACE

MARKUS FIRTZ
EX-EUTELSAT/SES

STEVE BOCHINGER
BCS

JAROSLAW JAROWSKI
EDGE AEROSPACE

ELENA GIGLIO
D-ORBIT

NIELS BUUS
EX-GOMSPACE

- ESPI MEMBER
- ESPI ADVISORY COUNCIL
- EXTERNAL EXPERT

The ESPI team whole-heartedly thanks all the participants for not shying away from an open and candid debate, for bringing their perspectives forward with poise and purpose, and for recognising the importance of this discussion at a time when Europe's space industry is being redefined through internal and external dynamics.

TAKING CONTROL OF THE LIGHT

THE WHALE

The whale embodies the balance of scale, speed, and vision required to navigate the depths of the world's oceans. Its form, a constellation of fish of varying sizes, reflects the living fabric of Europe's industrial and political ecosystem. Diverse yet interdependent, moving in concert to create a powerful whole. The whale may be blinded by the light, but it remains in motion, determined and unwilling to be stranded ashore. In time, it not only reaches the light; it learns to master it.

SPEED SCALE SKILL VISION

Strategic Levers to Reclaim Europe's Industrial Edge in Space

Schwarzenbergplatz 16
1010 Vienna, AT

office@espi.or.at
www.espi.or.at